

**On Experts and Algorithms:
Who Decides When AI Decides?**

By

Matya Stavnitzky

A thesis to be submitted to the
Cognitive Science Program
Mount Allison University
In partial fulfillment of the requirements for the
Bachelor of Arts degree with Honours

April 22, 2026

Thank you to my supervisor, Dr. A. Inkpen for being incredibly generous with your discussion, editing, and time. Thank you also to Dr. R. Moser, Dr. A. Hamilton-Wright, and Dr. D. Inkpen for your thoughtful insight.

Thank you to my housemates for filling every day with whimsy, to my friends at the library table for many laughs and conversations, and to my family for always being my strongest supporters.

Thank you as well to the many marvelous educators I have been privileged to learn from.

Table of Contents

<i>Abstract</i>	<i>1</i>
<i>Introduction</i>	<i>2</i>
<i>1. Should AI Help with Diagnosis and Treatment? Exploring AI in Medicine.....</i>	<i>8</i>
1.1 Reasoning Behind ML Models for Diagnosis	9
1.2 Reasoning for Deep Learning in Medicine	11
1.3 Bias as a challenge to AI in Medicine	12
1.4 Why we can still use Deep Learning in Diagnosis.....	17
1.5 The Shaky Road from Diagnosis to Treatment: Why Deep Learning Challenges Persist	18
1.6 Conclusion.....	21
<i>2. Should AI Advise Climate Scientists? Exploring Language Models in Scientific Assessment</i>	<i>23</i>
2.1 Background.....	23
2.2 Why LLMs Should be Used for Scoping Literature	27
2.3 Why LLMs Should Not Read.....	32
2.4 Why LLMs Should Not Write.....	38
2.5 Conclusion.....	46
<i>3. Conclusion: A Challenge to the AI Paradigm.....</i>	<i>49</i>
3.1 Challenging Assumptions of AI Predictions	50
3.1.1 Questioning if we Can Represent the World in Data	50
3.1.2 Questioning if the World Follows Stable Patterns that Will Persist.....	54
3.2 How is this Different from Science?	56
3.3 What is the Value of Human Expertise?	60
<i>Bibliography.....</i>	<i>63</i>

Abstract

With the growing capabilities of Artificial Intelligence (AI) models, we seem more inclined to treat these models as if they were experts. This thesis aims to argue that despite increased capacity in a variety of tasks, AI cannot replace scientific experts in decision-making. Chapter 1 focuses on AI in the medical context as a case study for whether AI can assist human expertise. It demonstrates that methodological similarities between reasoning in medical science and the generation of AI outputs are not enough to justify the usage of AI. Chapter 2 explores the application of LLMs to produce scientific assessments, specifically focusing on the Intergovernmental Panel on Climate Change (IPCC). This chapter further establishes that AI models diverge from scientific methodology because they focus on knowledge output rather than the process of its production. Chapter 3 pulls together ideas from the previous two chapters to make claims about the value of *human* decision-making. Demonstrating that assumptions underlying AI reliability depend on human values, but unlike science, the production of AI models does not address these values in clear, transparent, or communal ways.

Introduction

We rely upon experts or groups of experts to help make high-stakes decisions involving complex information. When we are unwell, we consult medical professionals; in the court system, expert testimony informs legal judgments; and in the face of environmental uncertainty, governments turn to scientific panels to guide policy. Because of this reliance, a substantial body of literature has examined the role of “expertise” in political, judicial, medical, and environmental decision-making (Collins, 2014; Kitcher, 2011; Moore, 2017; Nichols, 2024; Oppenheimer et al. 2019; Pielke Jr, 2007). Our reliance on experts, however, has recently been challenged by improvements in Artificial Intelligence (AI) systems, particularly their language capabilities, which make it easier to find and understand information. This has sparked renewed, and in some ways new, debates about the role of experts in informing our decision making. For instance, it has been suggested that AI, with its capacity to rapidly process and summarize large volumes of information, could streamline scientific assessment by automating stages such as screening and analysis (Al Khourdajie, 2025). Automating decision-making is not entirely new, of course, and statistically-based AI models capable of prediction and data synthesis have been used in decision-making for years, well before the resurgence of interest driven by large language models. In medicine for example, AI models have supported diagnosis since the 1950’s (Lusted, 1955; Göndöcs & Dörfler, 2024). Today, a number of AI models support clinical diagnoses by interpreting imaging tests (e.g., X-rays, MRIs, CT scans), blood tests, and biopsy results (Al-Antari, 2023). However, this growing reliance on AI raises unresolved questions about how its decision-making capabilities diverge from, or complement, human expertise. Accordingly, this thesis investigates three central questions: (i) under what circumstances is it reasonable or advisable for AI to assist human expertise? (ii) under what circumstances is it reasonable or advisable for AI to replace human expertise? and (iii) ultimately, what is the value of *human* decision-making?

My concerns in this thesis are primarily epistemic, that is, I want to know when AI is a reasonable substitute for expert judgment. For the sake of my argument, I will largely set aside questions about whether we should use AI based on the potential ethical, legal, and political costs of doing so – ranging from copyright infringement and labour

displacement to the existential threat of AI authoritarian control. My focus is motivated by the observation that, regardless of broader concerns, AI systems are often integrated into decision-making as soon as they appear capable, without sufficient critical evaluation of whether they should be used in these contexts. One example is the COMPAS recidivism prediction algorithm, which was adopted to advise judicial decisions on bail in the United States. Subsequent studies revealed serious issues of fairness, including racial and socioeconomic bias, as well as performance no better than laypeople with little to no relevant expertise (Dressel and Hany, 2018). This case illustrates that even as questions about the ethical development of AI remain urgent, we must concurrently engage in discussions about its appropriate use.

Complicating this discussion is the ambiguity of the term “AI,” which is used to describe a wide array of algorithms, including those for image classification, media generation, and recommendation. The label often functions as a marketing term, especially when it is advantageous to suggest that a technology possesses human-like capacities (Bender and Hanna, 2025). This is so significant that there have been outcroppings of “AI” that utilize humans to complete tasks they claim the technology is capable of. For example, Amazon’s “Just Walk Out” system was marketed as an AI powered cashier-less checkout but in practice it relied heavily on thousands of human reviewers to verify transactions (Bridle, 2024).

Even within academia, AI is less defined by shared methods or vocabularies than by a common goal. As benchmarks for “human capability,” such as chess playing or pathfinding, have been surpassed, new ones have been set, continually redefining what counts as AI. This shifting boundary means many algorithms originally developed in the AI field - such as breadth first search, depth first search, and A* - are no longer commonly referred to as AI.

In mainstream media the term “AI” often refers to generative models, models like language models (LLMs), that can generate high-quality text, images, and other content based on the data they were trained on. To avoid confusion, this thesis will use the term differently. Here the term *artificial intelligence (AI)* is used as a blanket term for all AI models. This includes deep learning and machine learning models. *Machine learning (ML)* refers here to models more directly grounded in statistics, such as naïve Bayes

classifiers and random forest models. These models tend to be trained on labeled data, and smaller datasets, for specific purposes. Alternatively, the term *deep learning*, will be used when referring to the models discussed by mainstream media. This includes the generative models' that mainstream discussions typically focus on, which are grouped together by their capabilities but are built with the same deep learning architecture; with neural nets that utilize backpropagation and a transformer-based model. Deep learning models become effective by "learning" patterns from large datasets; they are often developed for general applications instead of specific goals. The term *large language model (LLM)* will be used when referring specifically to text producing generative models.

The distinctions among these terms are not as clear as presented above. For instance, ML is a subfield of AI, and deep learning is a subfield of ML (IBM Data and AI Team, 2023). However, distinguishing between these terms will be useful so readers know roughly to which models I am referring. Especially because ML and deep learning utilize different methodologies - ML relies on statistical information from small data, while deep learning focuses on predictions from large datasets.¹ Some claims in this thesis apply to AI generally, others focus on a specific type of model.

This thesis is structured into three chapters. The first chapter focuses on AI in the medical context as a case study for whether AI can assist human expertise. Most AI models used by medical practitioners do not aim to replace human actors.² For instance, Göndöcs and Dörfler (2024), claim the role of AI algorithms in medicine is to produce accurate predictions which are then supplemented with judgements by human experts. In other words, medical practitioners use AI predictions as information to inform their decisions. Existing research suggests these AI-human hybrid approaches to diagnosis, in which AI supports rather than replaces experts, improve diagnostic outcomes (Zöller, 2025).

¹ This distinction is not cut and dry. Many people are working on methods to produce strong deep learning models on smaller datasets. For example, Li et al. (2023) focuses on removing redundant information from datasets, reducing the size while retaining relevant information. However, for the most part deep learning approaches require much more data than ML models to make strong predictions.

² With the exception of some recent models developed to do medical screening. For instance, a hospital in the UK has implemented an AI system to screen photos of suspicious moles for skin cancer to determine if a specialist should examine them (Hayes, 2025).

The first chapter will argue that although AI information can be trusted to inform some medical decisions, such as diagnosis, this is not the case for all medical decisions. I will first discuss why we trust ML to help evaluate medical images. After, I will argue that deep learning and ML have differing methodologies, making unresolved but recognizable challenges in medical imaging, such as identifying bias, more significant. This means we should question the application of deep learning models for diagnosis and treatment. I will argue that this is ultimately not an issue when using AI models to inform diagnosis. However, it is an issue for using AI models to inform treatment decisions. This is because while diagnostic decisions can rely on agreed upon standards for what constitutes a correct decision, treatment cannot. This chapter establishes that our trust in AI models requires assessment of what the correct decision is.

Chapter 2 explores the application of LLMs to produce scientific assessments, specifically focusing on the Intergovernmental Panel on Climate Change (IPCC). This is used as a case study for when AI can replace human expertise. Scientific assessments produce reports that summarize scientific findings for policymakers. LLMs can search through scientific sources and summarize them, so it is conceivable that these models could produce these scientific reports. However, I will argue that LLMs should only minimally automate scientific assessments – like in chapter 1, AI should be used as a tool to inform experts not as a replacement for experts. This will be argued for two reasons. The first is that inherent limitations of LLM models mean we should expect these models to perform poorly for scientific language. The second is that AI automation overwrites the scientific processes that are needed to produce useful knowledge.

The second chapter will start with a brief background on the IPCC. I will then argue that scientists can use LLMs to scope literature – because limitations of the model, such as bias, are addressed by scientific processes that aim at critical emendation. Conversely, I argue that scientists should not use LLMs to read scientific papers or write their reports. This is on the basis that when LLMs read scientific papers they introduce factual inaccuracies and overwrite valuable discussions about scientific terms. Similarly, when LLMs write scientific reports, they undermine required nuance as well as remove the process of deliberative acceptance – a process Beatty and Moore (2010) argue is crucial because it allows scientists to evaluate multimodal and discordant information and

values.^{3,4} This chapter emphasizes the role of value assessments in producing scientific knowledge, and that AI models overwrite these value assessments instead of replacing them.

Chapter 3 pulls together ideas from the previous two chapters to make claims about the value of *human* decision-making. Historically, digital systems have often been regarded as inferior to human judgment (Schaffer, 2019). However, with the improved performance of AI across a range of tasks, this assumption has shifted (Ferrario, 2024), prompting renewed debates about whether and when machines might rival or even surpass experts. Building on the previous chapters, this section demonstrates that AI reliability depends on how we assess human values in the production and use of these models. This chapter will start by demonstrating that our trust in AI information depends on value-judgements about the collected data, and persistence of patterns in the world. Then, because science must also contend with human values, the chapter will further distinguish AI development from scientific inquiry, arguing that while science (ideally) addresses values using clear, transparent, and communal processes, AI models opaquely introduce values. I will argue the conclusion we should draw from this is to accept AI when we can verify that our assumptions about data and patterns are well-founded. This value assessment of data and patterns must be done by human actors. Thus, the value of human decision making is that it communally and collaboratively assesses and agrees upon how values should inform our decisions.

Overall, this thesis aims to argue that despite increased capacity in a variety of tasks AI cannot replace humans in decision-making. In many decision-making contexts, there is no single, easily measurable metric for determining what counts as the ‘best’ outcome or decision. While straightforward performance measures - such as those used in medical imaging - can provide clear benchmarks, qualities such as fairness, and the integration of complex context-sensitive information resist simple quantification.

³ The IPCC aims to assess a wide variety of information, such as scientific, technical, and socioeconomic information. This evidence is “multi-modal” (i.e., comes from different disciplines) and often “discordant” (i.e., inconsistent or incomparable) (Beatty & Moore, 2010; Sarewitz, 2004; Stegenga, 2009).

⁴ “Values” here meaning standards by which we make decisions. These can be classified as epistemic (i.e. parsimony, evidence, predictive power) and non-epistemic (social, political, ethical).

Important decisions are not separate from our non-epistemic values. Thus, our decisions - and the information fueling them - must account for these underlying values.

1. Should AI Help with Diagnosis and Treatment? Exploring AI in Medicine

The goal of this chapter is to argue that although Artificial Intelligence (AI) systems can be epistemically aligned with some medical decisions, there are important reasons not to use AI information to inform all medical decisions. I will begin with medical imaging, where machine learning (ML) models already aid the assessments of radiologists (§1.1). I will argue, with a brief history, that ML models were adopted on the grounds that they (1) share a methodological similarity with methods humans use when determining a diagnosis, and (2) have high reliability. From this historical basis, I will then demonstrate that those in favour of wider employment of deep learning models seek justification with the same arguments of methodological similarity and reliability, but with greater emphasis on reliability (§1.2). However, contemporary deep learning AI models are not perfectly analogous to the older ML models used in medical imaging (§1.3). In fact, deep learning methods operate under a different paradigm than the ML ones. Whereas ML models rely on expert knowledge and theories to leverage data, deep learning models focus on generating reliable predictions from aggregated data. I will argue that the focus on aggregated data makes unsolved, but recognized, issues in medical imaging more significant for medical diagnosis. For example, the risk associated with bias is greater because it is harder to identify when bias is present in deep learning models.

The subsequent sections of this chapter (§1.4) will articulate why we might still think deep learning models could aid diagnostic decisions. But I will further contend that these arguments do *not* extend to helping practitioners determine treatment (§1.5). This is a significant conclusion because the ability of LLM models to produce human-like text has led to the impression that these models can fulfill tasks in a human-like capacity, including developing treatment plans. I will argue that in the context of determining treatment, unlike diagnosis, bias presents an epistemic challenge that cannot be solved. This means that AI models alone cannot provide the information needed for good medical decision-making. My conclusion (§1.6) will be that in some medical scenarios a data-based approach is not sufficient to determine what we should do; human judgement is needed. Treatment is such a scenario.

1.1 Reasoning Behind ML Models for Diagnosis

This section uses a historic account of ML models in medical imaging to understand the adoption of AI models into other medical contexts. I argue that the reasoning used to justify ML for medical imaging is also being used to justify AI models that aid diagnosis and treatment decisions.

Radiology is the field where AI systems have historically been the most prevalent. Research determining whether computers could help analyse radiographic images began in the mid 1950s (Lusted, 1955). In the 1960s and 1970s, researchers started to investigate automated detection and classification of abnormalities in applications such as breast images and chest radiographs (Giger et al., 2008). This includes what is cited to be the first application of ML algorithms in medical imaging – the use of Bayes rule by Gwilym S. Lodwick to calculate the probability of a bone tumor diagnosis (Lodwick, 1966). In 1998, the ImageChecker M1000, a computer system that identified and marked regions of interest on mammograms, was the first commercial system for computer aided detection (CAD) approved by the FDA (U.S. Food and Drug Administration, 1998). Other systems for mammography have since obtained FDA approval, and many these are being used clinically.⁵

The development and adoption of ML systems into clinical practices were justified by two main arguments: (1) AI systems use the same information and techniques that radiologists do, and (2) they improve the reliability of diagnosis, leading to the correct diagnosis more often. To illustrate this let us consider the paper by Lodwick (1966). Lodwick proposed Bayes theorem to calculate the probability of a diagnosis given features such as the location, size, and shape, of a tumor (*Ibid.*, p. 488). This is justified by claims that the radiologist and the algorithm use *the same* data and process to identify disease. The radiologist recognizes a lesion, determines important information, and then compares the data to the findings of the diseases she knows (*Ibid.*, pp. 486-7). The algorithm scans an image to find a lesion, extracts important information, and then compares the data to the probabilities of findings from diseases that are present in its

⁵ Commercial systems are proprietary, so little can be said about the algorithms used. However, computer vision has historically been the domain of AI, so it is reasonable to assume these CAD systems use AI methods.

training data. More specifically, the algorithm scans the image focusing on the edges of structures, which is evidenced to be similar to human scanning methods (*Ibid.*, p. 488). The data the algorithm uses is based on the data that has been digitally recorded by radiologists (*Ibid.*, p. 488) and is thereby the same data that radiologists consider. The difference between the algorithm and the radiologist is that the algorithm can be more reliable. Radiologists miss or misidentify a number of lesions, and the algorithm may reduce the number of false positives and false negatives (*Ibid.*, p. 486-7). Additionally, the algorithm can provide more accurate diagnosis because the radiologist may improperly weight the importance of certain traits when determining a diagnosis (*Ibid.*, p. 487).

As ML models have proven to be increasingly effective for aiding radiology it has been easier to propose models with methodologies that differ from the human approach. Especially if they continue to show good reliability. This is especially clear in the field of radiomics – a newly developing field that extracts invisible data such as texture from medical scans of tumors. This field is influenced by image-guided biopsies which indicate there are differences in how the proteins in tumors are spaced that are undetectable to the human eye (Lambin et al., 2012).⁶ Radiomics aims to take the data-driven methodology that has been accepted in medical diagnosis and expand it to features that are unobservable to the human eye.

Similarly, the success of deep learning in a variety of tasks has supported the adoption of these models for medical imaging. A growing number of physicians have favourable views of using AI in the clinical setting, and 32% use AI tools in their clinical practice (American Medical Association, 2025).⁷

⁶ This is not the only influence. I believe it has also been influenced by the increasing role computed images play in medical imaging. The shift from film based medical imaging to scans which collect data and require computer reconstruction to produce an image, make it feel more natural for computers to play a larger role in finding features in the images.

⁷ The term “AI” is used here because the source does not indicate what models’ people are referring to, but it is likely that the practitioners are referring to deep learning models.

1.2 Reasoning for Deep Learning in Medicine

Arguments supporting deep learning models in the medical context use the same methodology and reliability justifications used to support ML in medical imaging. This is an intuitive extension. Like medical imaging, medical diagnosis and treatment are understood to be data driven. Diagnosis aims to relate available information to medical conditions – these are often defined by disordered physiological processes. For example, a tuberculosis diagnosis is assessed by tuberculosis symptoms (e.g. coughing, chest pain, fever) and the presence of *Mycobacterium tuberculosis*. Doctors, and other medical experts consider a range of signs, symptoms, background information about patients, and results of diagnostic tests in order to infer whether a person has a particular disease (Stegenga, 2018, p. 175). In evidence-based medicine, a diagnosis is determined by comparing the symptoms (and defining characteristics, i.e. age) to meta-analyses.⁸ Similarly, the most effective treatment can be found by referencing meta-analyses. Meta-analyses combine data from independent studies to establish overall treatment effects, assess safety and risk, and resolve conflicting findings – they are often treated as a “default approach” to causal inference in evidence-based medicine.⁹

Support of evidence-based medicine provides the strongest argument for using deep learning models in medical diagnosis and treatments. The argument for deep learning models in medical diagnosis may go as follows. Like with medical images, deep learning AI models can use the same data practitioners consider when determining a diagnosis and treatment. For instance, AI models that aim at diagnosis can consider the same signs, symptoms, and background information as medical professionals. Richens et al. (2020) even suggest that – like physicians – AI models can determine the underlying causes of a patient’s symptoms, not just learn correlations between patient data and disease occurrences. Not only can AI systems use the same data that medical professionals consider, but they also have a high reliability. Recently, Zhao et al. (2026) presented an AI system called DeepRare which was more accurate in providing potential

⁸ In instances where the result is still ambiguous, practitioners may conduct additional medical tests.

⁹ Meta-analyses have been critiqued for only proving “an event has occurred somewhere” instead of showing that the event will happen in a specific circumstance (Cartwright, 2011). For instance, a meta-analysis may only demonstrate that a cancer treatment has been effective once, instead of demonstrating that a cancer treatment will be effective each time it is applied. However, metaanalysis are still considered the gold standard for evidence-based medicine.

diagnosis of rare diseases than physicians with access to a search engine. This suggests that the ability of AI systems to leverage medical data can improve diagnosis and should therefore be used in the medical system.

However, this reasoning depends on a successful analogy between ML for medical imaging and current AI models, and these models are not the same. ML models have methodological practices similar to statistics – where expert knowledge and theories are used to leverage data – but the success of deep learning, specifically with LLMs, has shifted the focus to generalized models that work by training on aggregated data. There is also more push for automation. Because LLMs are able to produce human-like text there is greater belief in their ability to fill the role of a human. These differences mean that models today have a greater risk of bias.

1.3 Bias as a challenge to AI in Medicine

The next few sections will discuss how deep learning models trained on aggregated data make issues of bias worse.¹⁰ But it is useful to sketch out my argument for the rest of this chapter so that its flow through these sections has better context. In this section (§1.3), I show that AI models (both ML and deep learning) will always underperform for some subset of the data – this means they *will always* be biased. However, bias is not always a reason to avoid using AI models. When taking a more statistical approach, bias is less significant because the focus on good data quality makes bias more likely to be identified. In §1.4, I argue that bias is of greater concern for deep learning models that train on aggregated data because it is harder to identify. But even so, bias is still not always enough to claim we should not use these AI models. In medical diagnosis the data that should and should not be used to make decisions is well defined by pre-existing diagnostic criteria; making it easier to recognize when bias may be present. In §1.5, I argue that this is not to say that bias is never an issue for the adoption of AI into medical contexts. In determining medical treatment, the issue of bias means that AI should not be adopted.

¹⁰ Bias is not the only concern with adopting newer (and older) AI models. There are also for example, issues with data privacy that arise from aggregated data and deep learning. However, these issues tend to lead to legal issues rather than epistemic ones.

Note on the Term Bias

The term *bias* typically refers to a prejudice for or against a person, group, thing, or idea, often in a way that is considered unfair or unreasoned. It often has a negative connotation. I will refer to this type of bias using the term *societal bias*. When I talk about AI models being biased, I mean that the algorithms produce a different (typically less effective) set of predictions for a subset of people. This relates to societal bias when it follows patterns of oppression but does not necessarily have to be harmful. Bias is intended as a neutral term that refers to having a particular perspective of the world.

Bias in AI models is not a new issue. Concerns about racial bias were raised in 2018 by Joy Buolamwini who established that facial recognition systems performed worse for black women than other groups. Even in more statistically based machine learning models for medical imaging there have been concerns about bias. For example, AI that analyse x-rays have been found to have gender and racial biases (Trafton, 2024). In fact, bias is such a prevalent issue in AI models that many researchers are working to prevent it. But bias itself is not a solvable problem, in AI models there will always be some subset of the data for which a model performs worse. As so many researchers are working to reduce or remove bias, this is a large claim and may be unintuitive. It stems from the fact that data is rarely free from noise, meaning not all data follows patterns perfectly. Therefore, to make AI models more accurate developers have to be careful to train the model to predict the pattern not the noise. This means more accurate models will ignore outliers – and thus AI models perform worse for outliers than they do for data that follows the predicted pattern.

To illustrate why bias is inherent to AI models let us consider the problem of overfitting a linear regression model. Linear regression is a simple ML method that fits a straight line to data. Typically, the data is split into training and testing data.¹¹ We start

¹¹ This is for supervised learning, where the data is labeled (meaning the data identifies the outcome the model should be able to produce). LLMs often use a self-supervised method where the correct outcome is inferred from the data.

with the training data in figure 1. This is the information we will use to train our ML model.

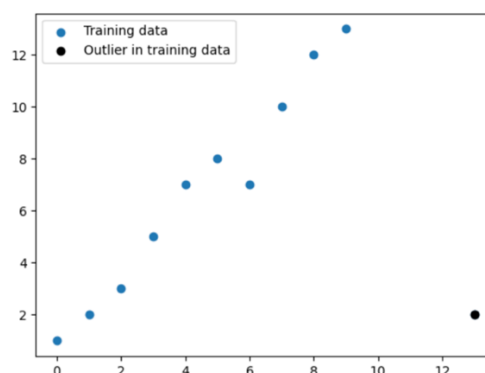


Figure 1.

To fit a line to the training data the simplest approach is to try to minimize the distance from the line to all the points. However, as is indicated on the graph one of the points is an outlier, it doesn't follow the pattern of the data. We may want to ignore this datapoint to make a better model. Figure 2 shows models that include and exclude the outlier.

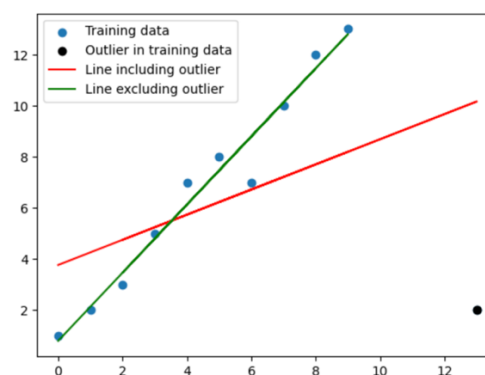


Figure 2.

From here we have produced 2 potential AI models. It appears that the green line, from the model which excludes the outlier, does a better job of predicting the data. To evaluate how this model performs, we can use the testing data – which is data the model has not been trained on. Dividing the available data into testing and training data allows

us to test if our models are just perfectly predicting the information they start with or if the models are capturing larger patterns in the data.¹² Figure 3 shows the rest of the data.

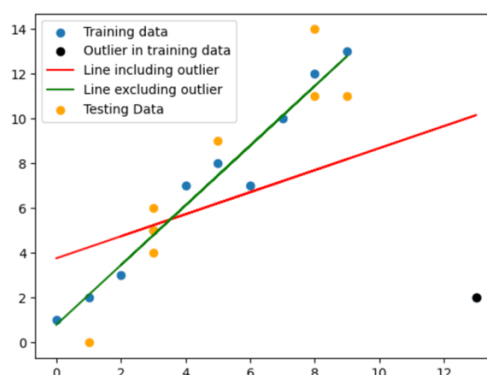


Figure 3.

From figure 3 we can see that even in our training data the outlier does not follow the pattern. In this instance ignoring the outlier allows us to make better predictions. The green line is much closer to the rest of the data than the red line. This is a simple example, but it aims to provide some justification behind the idea that increased accuracy in general requires worse accuracy for some datapoints (given that noise or outliers are almost always present). In this way, AI algorithms ignore data that does not fit expected patterns. It is not just that AI algorithms *can* underperform for some data, good AI algorithms *must* underperform for some data. More complex algorithms aim to predict the data in more sophisticated ways than a line, but they still must balance prediction for their current data with the expectation of what is common in the data. AI algorithms predict accurately for data that fits the average and the further you get from the expectations of the data the worse the algorithm is at prediction (Peterson, 2024).

Bias in AI is worrying when it is influenced by societal biases. This is seen in medical imaging. For instance, Gastouniotti et al. (2022) found an AI model for assessing the risk of breast cancer was less accurate for Black women. Societal bias is an issue the field must contend with and actively works to address. However, it has not been enough to reject the use of AI for interpreting medical images. This is at least in part because of oversights caused by a racist society. AI models often share biases already present in the

¹² The technical term for the idea explained here is overfitting.

medical system. Radiologists, like algorithms, are more likely to have poor outcomes for black women (Farr, 2024). However, I believe the type of data used in medical imaging means AI bias is not so significant. Radiology is one of the most data-informed disciplines in medicine (Erickson & Bartholmai, 2002). Radiologists are trained by being exposed to a large number of images of the pathologies they aim to identify. Further, the applications where AI models are most common, such as for identifying cancer tumors, have relatively few ways they manifest. There tend to be fewer outliers, because the pathologies being identified follow strong patterns (Avanzo et al., 2024). This means we can expect relatively few biases caused by variations in the ways a pathology manifests. Thus, if we can remove societal biases that are currently present in datasets, such as racial bias, we can expect reliable outcomes from AI algorithms.

But can we remove societal bias currently present in datasets? While it is not possible to remove all societal bias in datasets, it is possible to mitigate the societal bias that we are aware of. In the case of medical imaging, we can identify societal bias by comparing the performance of the AI system among different groups of people; and we can remove societal bias by ensuring the data is representative of the population and does not make marginalized communities outliers. It thus appears at least possible to mitigate societal bias, such as racial bias, in AI applications that examine medical images. For example, a model that aims to assess ductal carcinoma in situ or invasive breast cancer was found to have no racial bias (Giouroukou et al., 2026).

However, societal bias does not only occur because of underrepresentation in datasets. It can also occur because of misrepresentation in datasets. That is, when datasets do not represent the correct patterns. In the case of medical imaging this may be the mislabelling of a tumor. For diagnosis, it may be a wrong diagnosis, or no diagnosis when a medical condition is in fact present. In treatment, it may be when the wrong treatment is given, or when treatment is incorrectly withheld. In medical imaging, misrepresentation is not a significant concern because it is possible to track the actual outcomes of a diagnosis with other factors, such as completing a biopsy on a suspected tumor. That is, the data can be connected to other information that verifies its

truthfulness. This means identifying incorrect data is not terribly difficult, although whether it is done in practice is a separate issue.

In addition, there is an argument to be made that in this instance bias in AI models is easier to identify than bias in humans. This is because AI models are centralized, meaning it is easier to collect data on their predictions, making it easier to assess for accuracy. There is only a need to identify bias in one source. Whereas identifying bias from radiologists means collecting information from many radiologists, which may be distributed between many different types of provider systems. Similarly, correcting bias in AI models may be easier, again because the system is already centralized. Changes only have to be made to the AI model, as opposed to training many radiologists. While both radiologists and AI models should aim to produce less societal biases, the ease at which we can identify and address societal bias in AI models may help us justify their use.

1.4 Why we can still use Deep Learning in Diagnosis

Above I have argued that bias in ML models does not limit their application to medical imaging. Here I will extend these arguments to deep learning that aids medical diagnosis.

Like medical imaging, a diagnostic decision is often supported by other information, such as specific tests for specific medical conditions, that verify its truthfulness. Diagnosed conditions are often delineated with a set of symptoms that indicate a condition is present. This means the information that should be relevant to a diagnosis is well defined, it should be the relevant symptoms. Evidence-based medicine identifies the bounds of diseases, as well as documenting a variety of ways they may manifest. This has already been completed before any diagnoses are made. This is important because it indicates there is an agreed upon answer to diagnostic decisions, that we can use to assess if diagnostic decisions are correct. Like with identifying a misidentified tumor, we may identify a misdiagnosis by conducting an additional test, or by re-examining the symptoms in accordance with diagnostic guides.

Unfortunately, it is harder to identify issues that arise from a lack of diagnosis. It is difficult to determine what is absent, especially if someone without a diagnosis never

returns to have it re-checked.¹³ We know that lack of diagnosis, like misdiagnosis, follows societal patterns of oppression. Black patients are less likely to be believed when they say they are in pain, and less likely to receive appropriate diagnosis from medical professionals (Schoenthaler & Williams, 2022). In this instance we may again use an argument from medical imaging. We may say that the type of AI bias that occurs from a missed diagnosis will be the type of bias that medical professionals are already making, and it is easier to identify this in an AI system than in a large group of medical professionals. So, AI models used for diagnosis may actually allow us to better identify when bias is present.

1.5 The Shaky Road from Diagnosis to Treatment: Why Deep Learning Challenges Persist

Unlike the previous cases, in treatment it is difficult to determine what the correct decision is. While meta-analyses may provide the most effective treatment overall, this may not always be the best treatment for an individual. We may expect medical conditions to follow more regular patterns because of the physiology of our body.¹⁴ Whereas the efficacy of treatments may vary widely depending on a number of social factors. For instance, the success of a treatment may be affected by the ability/propensity of a person to access a treatment, their overall nutrition, and other lifestyle decisions. AI models are only as effective as the world ‘underneath’ is stable – they rely on the assumption that there are patterns to predict.¹⁵

However, there are many factors that affect the efficacy of treatment, and it is not clear that all of these follow predictable patterns. Consider tuberculosis, despite being preventable and curable since the 1950s, tuberculosis persists because of economic inequality, stigma, and a lack of political will (Green, 2025). Many people cannot access the more effective tuberculosis treatments that are available because they are more

¹³ Or challenge the lack of diagnosis in other ways, such as by challenging the practitioner. Which, unfortunately, may at times be necessary.

¹⁴ Although when it comes to who has medical conditions this is not strictly true. Health and illness follow a social gradient, with lower socioeconomic status leading to worse health outcomes (WHO, n.d.). However, from the perspective of a doctor making a diagnostic decision, the information that must be considered is primarily the way a medical condition manifests in the body.

¹⁵ More on this in chapter 3.

expensive. Instead, less effective treatments are used, often involving a medley of drugs that must be taken regularly. This medley is ineffective for some because they are unable to take the drugs at the specific times. It is ineffective for others because they do not have the nutrition required for the drugs to be effective. Still others may refuse the drugs because they do not trust the medical system that is providing them. This example illustrates how difficult treatment decisions can be: they must be based on the type of treatment a person is willing and able to accept, as well as the treatment that is available. Even if a meta-analysis indicates a treatment is the most effective, this is not always the one doctors decide to use. That decision must be influenced by factors beyond the efficacy of treatment.

Additionally, it is difficult to determine that providing a particular treatment was the correct or incorrect decision. It is possible that the treatment applied was correct, but the patient still had a poor health outcome. Or that an ineffective treatment was provided, but the person survived the natural course of the illness anyways. It is much harder to assess the correctness of a treatment, than it is to assess a diagnostic decision. Medical practitioners are ethically bound to provide the treatment they believe to be most effective, and it is difficult to determine what the outcome of a different decision would have been. This makes bias harder to identify because we rely on identifying when predictions are wrong to identify bias.

What's more, it is difficult to determine what information should or should not be included in the decision-making process. In both medical imaging and diagnosis, the data that should be considered is well defined; we should consider medical images, and symptoms respectively.¹⁶ However, a broad range of factors can affect the success of treatment decisions. Care professionals may consider factors such as lifestyle, socio-economic status, etc., that change how well a treatment can be followed and how effective it might be. For example, a doctor treating an elderly patient with type 2 diabetes must consider physiological factors such as blood sugar levels, as well as the patient's fixed income, which might make healthy groceries or expensive insulin hard to

¹⁶ Diagnosis is not quite as simple as presented here; we may also have to decide about characteristics of the person such as age and sex. But within even those decisions there are generally guidelines about how pathologies manifest in these difference reference classes. So, there is still some standardized way to assess the accuracy of a decision.

afford. They also evaluate the patient's physical dexterity for administering injections and their proximity to a pharmacy, to ensure the treatment plan is medically sound in the patient's daily life. That doctors must consider such a wide range of factors suggests that the data to include in AI systems for determining treatment should be very broad, which would make bias harder to identify. Especially because the data that leads to the most accurate predictions may be based on societal factors which we do not actually want to influence our medical decisions. Some of these biases may be easy to identify, such as racially biased predictions, however some of these predictions may appear to be accurate when they are contingent on social practices remaining the same.

To illustrate what that means, let us examine an early AI model that was used to predict which pneumonia cases would be the most severe. The model, once trained, indicated that people with asthma were less likely to die from pneumonia (Szabo, 2019). This is true, but only because people with asthma are more likely to take pneumonia seriously because we they know they are at higher risk. In this instance, the AI model picked up on accurate patterns in the data. However, these patterns exist because of psychological and societal practices, not because of biological processes or because of the effectiveness of treatment. In this way an AI model may be accurate but not useful for decision making. There is no reason to trust AI 'judgement' on a diagnosis beyond that it has historically been correct.

Further, the patterns of oppression in treatment are more prevalent than in medical imaging or diagnosis. This is in part because there are a larger range of societal factors that impact treatment decisions and outcomes. This means that even when we try to remove bias (i.e. by ensuring marginalized people are not outliers and that data has the correct decision outcomes) we still get predictions that produce bias. This is because the most accurate predication for treatment is one that is based in the prejudices and oppression that societies continually reinforce. The prevalence of bias has been seen in the use of LLMs for therapy. Bouguettaya et al. (2025) found that LLMS exhibit bias when dealing with African American patients, sometimes recommending radically different treatments for the same psychiatric illness.

In determining treatment – while we can determine a desired outcome – it is difficult to determine what decisions will lead to that outcome, and what data should

inform that decision. Right now, we (mostly) trust medical professionals to undertake this decision-making process. In other cases, it seems possible to trust AI systems to help with our decision making. However, in these other cases there is reason to believe there is data that can accurately determine what the decision should be. There are agreed upon standards for the evidence that experts use to make diagnoses, and there are ways to test the correctness of these decisions. This is not the case for treatment. There is no agreed upon assessment about what factors must be considered to determine the right treatment, and it is difficult to assess the effect of decisions. Even if AI predictions appear to accurately predict the world, this could be the case for a number of reasons, including ones we do not want to affect our treatment decisions. Because of this, the burden must be on AI developers to demonstrate that the data is well suited to inform the decision being made.

1.6 Conclusion

This chapter has discussed the adoption of AI models into medical contexts, focusing on medical imaging, diagnosis, and treatment. While all three applications appear to be data based, the nature of the data is different. Medical imaging provides the best data for AI adoption, with images that can be labeled by a variety of experts, providing a reliable ground for trustworthiness and accuracy. We can expect bias in medical imaging to arise primarily from poor representation, where outliers create bias. For medical diagnosis, we can expect more errors from incorrect labelling, as it is easier to misdiagnose or fail to diagnose and harder to verify when this occurs. However, even in this case the medical system provides reliable ways to assess a diagnosis, and standard criteria for how a medical condition may present. Determining the correct treatment is much more difficult, however. Selecting an appropriate treatment depends on a variety of social factors such as religious beliefs, lifestyle, patient goals of care, etc. It is difficult to determine what information should be considered, and how it should affect the final decision. There are no standardized ways to measure this, and determining the outcome of alternative treatments is often not possible. This means bias is difficult to assess, because there is no way to guarantee the data leads to the best decisions. Doctors may not even have an idea of what the best decision is.

I have argued that AI in medical imaging was originally adopted for two main reasons, methodological similarity, and reliability. Both of these arguments depend on the type of data an AI model is trained on. When there is an identifiable right answer, and when the data follows predictable patterns, AI models are well suited to the context. However, in the case of treatment there is not always an obvious right answer, and there is no guarantee that the predictable patterns are meaningful. That is, there is no reason to believe AI is well suited for the task. If we want to adopt AI for this purpose the onus remains on AI developers to justify how the data could lead to reliable predictions.

This chapter has thus far left the question of automation aside. Coarsely, there are two major ways AI can be integrated into diagnostic processes: it may focus on aiding human assessment, or it may aim to replace human assessment. This is a common question when it comes to the integration of AI systems, how much should AI automate? Should AI aim to augment human work or replace human actors? This question is especially relevant because increased AI automation leads to different epistemic challenges and goals for diagnoses.

As previously discussed, AI models that have already been integrated into the medical system have aimed to support human assessment, not replace it. It has been suggested that AI integration breaks the process of medical decision making into two components: a prediction and a judgement (Göndöcs & Dörfler, 2024). However, with advancements in AI capabilities it is not difficult to imagine an AI system could now function in the role of a human doctor. I do not believe this is the case – given what I have described above as the decision-making realities for doctors.

The next chapter will examine how different epistemic challenges arise when AI automates tasks instead of aiding them. This discussion will focus on the role of AI in scientific assessments about climate change. Where higher levels of AI automation have been more seriously considered.

2. Should AI Advise Climate Scientists? Exploring Language Models in Scientific Assessment

In Chapter 1, I explored how Artificial Intelligence (AI) models should be treated in the medical context as a case study for when scientific experts should use AI models to inform their decisions. This chapter examines if AI models can replace scientific experts, focusing on the application of AI, specifically large language models¹⁷ (LLMs), in producing scientific assessments. These assessments synthesize extensive bodies of scientific literature and are often designed to inform political decision-making. As a case study of this process, I consider the Intergovernmental Panel on Climate Change (IPCC). Although assessments are produced across many scientific fields, the arguments in this chapter are meant to apply broadly, as the core processes involved in creating such assessments remain largely consistent across contexts.

The arguments I offer in this chapter do not hinge on the current capabilities of LLMs, but on their intrinsic limitations. This distinction matters because developers are continually working to mitigate many present-day issues with LLMs. My goal, however, is to explore the applications of LLMs in scientific assessments now and in the future. Accordingly, I focus on limitations that cannot be overcome through technological progress, using the climate context as my primary example. I argue that while LLMs can serve as helpful tools for searching scientific literature, they should not be relied upon to read scientific papers or to write scientific assessments.

2.1 Background

The IPCC is the United Nations body tasked with assessing the science of climate change. It was created in 1988 amid rising public concern, spurred in part by NASA scientist James Hansen's testimony to the U.S. Senate that human activity was driving global temperature increases (Briggle, 2024; Conway & Oreskes, 2010). The IPCC is composed of a variety of scientific experts from around the world with the goal of

¹⁷ A quick note on terminology: An AI *model* is the method used to create outputs from the data, for example a large language model (LLM) or random forests (which is a type of ML classifier). An AI *system* is a particular implementation of a model (or a set of models) that aims to accomplish a specific task, i.e. ChatGPT. A model can be discussed from a more epistemic (theoretical) standpoint, while a system has more concrete goals and limitations.

“assess[ing] on a comprehensive, objective, open and transparent basis the scientific, technical and socio-economic information relevant to understanding the scientific basis of risk of human-induced climate change, its potential impacts and options for adaptation and mitigation.” (IPCC, 2013a). In short, the main goal of the IPCC is to summarize the vast scientific literature on climate change into understandable documents that can aid political decision making.

The reports the IPCC produces are used to inform climate policy, primarily through the UN Climate Conference – the UN Framework Convention on Climate Change (UNFCCC) (Briggle, 2024; IPCC, 2025a). At the UNFCCC, IPCC reports are used to create policy recommendations and non-binding plans for the reduction of greenhouse gas emissions (Briggle, 2024). The first policy the UNFCCC implemented was the Kyoto Protocol, which was later superseded by the Paris Agreement; these policies are multinational commitments to stopping climate change.¹⁸

The IPCC itself aims to be neutral on climate policy. In the Principles Governing IPCC Work they state:

“IPCC reports should be neutral with respect to policy, although they may need to deal objectively with scientific, technical and socio-economic factors relevant to the application of particular policies” (IPCC, 2013a).

This means that the IPCC does not advocate for policies, rather their reports outline the possible impacts of various policies. Because the IPCC aims to be policy neutral while dealing with literature and communication that involves non-epistemic value judgements (e.g., it’s hard to assess the risk of sea-level rise, and it’s possible mitigation, without attending to the risks it poses to specific coast-dwelling humans), they have sociopolitical processes which account for when value judgements occur (Briggle, 2024).¹⁹

The IPCC is composed of three working groups and a task force. The working groups produce reports on (1) the physical science basis, (2) impacts, adaptation and vulnerability, and (3) the mitigation of climate change (IPCC, 2025a). The Task Force on National Greenhouse Gas Inventories (TFI) is responsible for determining the agreed

¹⁸ That have not historically been followed very well. Which is unfortunate, but not relevant.

¹⁹ These include diverse author selection, extensive peer review, a conflict-of-interest policy, and specific guidelines for handling uncertainty.

upon methodology for calculating greenhouse gas emissions and removal by signatories to the Paris Agreement (IPCC, 2025c). Every five to seven years the IPCC produces a synthesis report based on the findings of the working groups and the TFI. The synthesis report integrates information from all the other reports and summarizes it for policy makers (IPCC, 2025b).

The question I tackle in this chapter concerns the proper role of AI in the production of current and future IPCC reports. Should AI be used? How should it be used? And what concerns should we be thinking about in answering these questions? Recently, Alaa Al Khourdajie (2025), a science-policy expert and Senior Scientist for the IPCC, has argued that AI can help address challenges the IPCC faces today, if it is integrated alongside human experts instead of replacing them. He argues that the ability of AI models to search through, group, and summarize huge amounts of information can help the IPCC synthesize more literature, and produce shorter reports; issues they have struggled with due to the exponentially growing climate literature.

The IPCC itself also recognizes the potential benefits of AI. In 2024, the IPCC posted and subsequently filled an Artificial Intelligence Support Officer position for Working group II, the group which focuses on impacts, adaptation, and vulnerability (IPCC, 2024b). Although the intended usage of AI is only vaguely described, it is suggested that AI could be used to: automate publication searching, process review comments, check consistency of key results, cluster research articles, and produce topographic maps (IPCC, 2024b). In the job posting, the IPCC also indicates that they intend to produce custom machine learning (ML) algorithms specifically for producing reports, although they do not indicate exactly what this means.

Al Khourdajie (2025) argues that ML models should be integrated according to pre-existing workflows. He suggests that early steps, such as screening and selecting sources, can be addressed with ML text classification, while models, such as living maps and databases, are more useful for reporting and visualization. This is valuable as it emphasizes that concerns about integrating AI models depend on both the type of model and why the model is being used. Limitations of AI can stem from models, or it can stem from how they are applied.

A different approach is taken for LLMs. Al Khourdajie argues that there are three inherent limitations in LLMs that are relevant to the IPCC: (1) propagation and amplification of bias, (2) factual inaccuracies, and (3) the erosion of nuance and calibrated uncertainty.²⁰ Al Khourdajie argues that the propagation and amplification of bias is inherent to LLMs because “LLMs are trained on vast amounts of text data, inevitably inheriting societal biases related to geography, language, gender, or perspective embedded within that data” (Al Khourdajie, 2025, p. 7). The other limitations—factual inaccuracies and the loss of nuance and calibrated uncertainty—stem from the structure of LLMs themselves. These models generate text by stochastically selecting outputs from a probability distribution learned from static training data. The distinction between addressable and inherent limitations is important to Al Khourdajie because different parties are responsible for the different types of limitations. Users of LLMs are responsible for utilizing these systems in ways that recognize their inherent limitations, while producers are responsible for making models that reduce addressable limitations.

While Al Khourdajie (2025) provides a systematic framework for applying many ML technologies into IPCC assessments, the guidance for LLMs is less specific. He recommends considering the inherent and addressable limitations of LLM models, but there are no strong claims about when these limitations mean LLMs should or should not be used. But we need such guidance with regards to LLMs as much as we need it with ML models, and as the discussion below shows, there are significant reasons specific to the IPCC reports and its context for why this is so.

The rest of this chapter, then, aims to demonstrate how the inherent limitations of LLMs, as outlined by Al Khourdajie, should constrain the usage of LLMs in IPCC reports. I argue that LLMs are acceptable for scoping literature but should not be used in high-automation contexts (such as reading papers, synthesizing literature, and communicating findings).

²⁰ I will discuss all of these limitations in further detail, later in this chapter.

2.2 Why LLMs Should be Used for Scoping Literature

I will argue that LLMs should be used by the IPCC to find relevant sources of evidence. This is for two reasons. First, the inherent limitations in LLMs—the erosion of nuance, factual inaccuracies, and the introduction of bias—are mitigated when LLMs only provide sources, and do not automate reading and writing. Second, I will further argue that LLMs can benefit the IPCC by increasing the scope of literature the IPCC considers, thereby reducing the overall (and already recognized) bias in IPCC assessments.

My first argument, that inherent limitations of LLMs are mitigated when they only provide sources, will focus on how the scientific process addresses LLM bias. I argue that LLM bias is mitigated through scientific processes that the IPCC already uses to mitigate individual human biases. The erosion of nuance and factual inaccuracies will not be discussed in depth because they are mitigated by limiting how LLMs are used. The erosion of nuance is a concern primarily when LLMs write, whereas factual inaccuracies are the most significant when people are reading LLM outputs. As discussed in Chapter 1, some degree of bias in AI models is unavoidable – it will always be possible to define a subset of the data upon which an AI model performs worse, unless all the data is homogenous or it follows a perfect pattern. For the IPCC, bias depends on what scientific information is assessed and how conflicting findings are resolved. Conflicting findings will be addressed later in this chapter; for now, let us consider how the IPCC currently decides which papers to include in their assessments.

The IPCC relies on selected authors, i.e., “experts,” to know or discover relevant scientific literature. For example, published information about how the IPCC selects its sources leads immediately to information about how the organization selects its authors. And IPCC guidelines specify that authors should consider a wide range of evidence. In Appendix A to the Principles Governing IPCC Work, the IPCC states,

“In Assessment Reports, Synthesis Reports, and Special Reports, Coordinating Lead Authors (CLAs), Lead Authors (LAs), and Review Editors (REs) of chapter teams are required to consider the range of scientific, technical and socio-economic views, expressed in balanced assessments.” (IPCC, 2013b, p. 4)

In the same document, under Annex 2: Procedure on the Use of Literature in IPCC Reports, it is further specified that,

“The author team is required to critically assess information they would like to include from any source. Each chapter team should review the quality and validity of each source before incorporating information into an IPCC Report. Authors who wish to include information that is not publicly or commercially available are required to send the full reference and a copy, preferably electronically, to the relevant Technical Support Unit. For any source written in a language other than English, an executive summary or abstract in English is required” (IPCC, 2013b, p. 17).

IPCC authors are expected to justify and document the inclusion of sources, especially when these sources are not published scientific studies. However, the IPCC does not, as far as I can ascertain, have specific procedures for how authors should search through the scientific literature. It is unclear exactly which processes authors use to find sources for the assessments. It seems reasonable to believe they rely on well-known scientific journals and common search engines, such as Google Scholar, and databases, such as Web of Science, to find their sources.

While the IPCC aims to provide an overview of the total literature of climate science, the size of the literature means it is not possible to consider all sources. Based on climate change studies indexed in the Web of Science, the peer-reviewed literature on climate change has ballooned, doubling roughly every 5 years since 1990 (Ford et al., 2025, p. 3). Meanwhile, the ratio of literature cited in IPCC reports has fallen from over 60% in the second assessment report to around 15% in the sixth assessment report (Ford et al., 2025, p. 3). Each IPCC report is longer than the last, but these reports still capture less of the total literature. This means that while the IPCC aims to be comprehensive in its assessments, it still must decide which information to include and exclude.

The IPCC has been critiqued for being Eurocentric and privileging Western scientific expertise, erasing more “holistic, pluralistic, and co-produced approaches to knowledge” (Wagner, 2025). In the IPCC’s sixth assessment report 99.95% of sources were written in English (Chavelli & Connors, 2023, p. 7), with publications from the United States and the United Kingdom accounting for roughly 75% of all references

(Chavelli & Connors, 2023, p. 8).²¹ It seems quite plausible that the lack of official procedures and processes for finding sources is at least in part responsible for these biases. The IPCC seems to assume that finding sources is a standardized practice for scientific experts as opposed to an ongoing process of determination within the scientific community.²² While the review process may help mitigate this bias, it is difficult to assess holes in a search, especially if there is no standardized procedure for conducting the search or if reviewers share the same assumptions, background, and values as those writing the report (Longino, 1990). The IPCCs literature review stands in stark contrast to practices in medial fields surrounding systematic review, where the gold standard requires clearly documenting and specifying search strategies and detailing the use of automation tools (Page et al. 2021).

Developing procedures around the use of AI may prompt the IPCC to establish clearer guidelines for literature scoping practices in general. The medical field offers valuable guidance on scoping practices both with and without AI. Notably, Cochrane - an organization renowned for producing high-quality, trusted systematic reviews, considered the gold standard in evidence-based healthcare - recently released a positional statement on the responsible use of AI in evidence synthesis. This statement outlines key procedures, including the requirement for transparent disclosure whenever AI systems are used in systematic reviews. It emphasizes that AI should be implemented responsibly, with human oversight, accountability, and rigorous methodological standards upheld throughout the review process (Flemyng et al., 2025).

There are reasons to think that using LLMs in this context, that is, for scoping literature, will be beneficial because it may mitigate some of the biases discussed above, if those biases result from a failure to find alternative, relevant sources. In the context of large literature reviews, the ability of AI to investigate a wider range of sources may improve the overall breadth the IPCC is able to achieve; similarly to how search engines

²¹ When assessing these statistics hubs of climate science expertise must be considered, but even still there is a bias towards expertise from these countries. Thank you to Dr. D. Inkpen for this comment.

²² Please note that the IPCC does not only find sources, but they also assess the reliability (or their trust in) those sources. However, this section specifically focuses on finding relevant sources within the vast literature that is published.

improved information access.²³ AI models could help flag relevant sources, to later be considered in more depth by human reviewers. If AI allows us to examine more sources, it would reduce the amount of relevant material that is not considered (false negatives). The trade-off is that there would also be some increase in irrelevant material that is considered (false positives). However, these would ideally be assessed and removed by human experts later in the assessment process. So, here, we are imagining AI as a source-finding tool.

As a toy example, imagine there are 100 total sources on climate change. Human reviewers can find 15 of these independently. An LLM can find 20 more sources, but 5 of these are not actually useful sources, they are false positives. The LLM misses 70 of the climate change sources these are false negatives – keep in mind humans were also not able to find these sources and missed even more than the LLM did. When the reviewers assess the sources provided by the LLM, they would ideally identify the 5 that are not useful. As a result, the final report would include 30 sources instead of 15. In this way an LLM could increase the total sources considered, while not introducing false sources. As AI systems are better equipped to scour large databases and group information than humans, this application seems to match the epistemic capabilities of both groups, especially considering the extensive (human) review processes that happen later.²⁴

One worry might be that AI would introduce its own biases into the literature search process. As we have said above (and in Chapter 1), it is an inherent limitation of LLMs that they are biased. This can be offset (somewhat) by allowing AI biases to be corrected by the same iterative process of science that is already in place to correct human bias. As the philosopher Helen Longino (1990) has argued, scientific knowledge is only objective and trustworthy insofar as it is a result of an iterative process of critical emendation, with peer review as a central component. As we saw above regarding biases in the current IPCC report, this is never perfect. But the same iterative review process in place to correct human biases would be applied to AI literature searches.

²³ This is especially relevant as search engines, like LLMs, introduce bias into the process of finding sources. White men tend to be the most commonly cited online, with marginalized groups underrepresented to varying degrees (Hawkins, 2021).

²⁴ This is truly intended only as a toy example, especially considering the huge size of the literature referenced previously. Expertise certainly has a role to play, but better search systems here might help lead experts to relevant papers that may have otherwise remained unfound in databases.

Furthermore, this adoption of AI to help search for resources is justified by the overlap in search engines and AI models. While we may think of semantically searching through sources as an AI task, many popular search engines already use semantic search and have been doing so since well before the rise of LLMs. Today LLMs and search engines have increasing amount of overlap in features.²⁵ Many LLMs can search the internet and provide you links to specific sources like a search engine would, and many search engines include LLM summaries in search results. If we constrain the usage of LLMs to finding sources, we may be able to consider LLMs to be alternative interfaces for searching the internet rather than an entirely new technology.²⁶ Especially, again, if scientists are still reading and considering all the sources once they are discovered.

Ultimately, this section has aimed to convince readers that LLMs could benefit the IPCC if they are used to find relevant sources.²⁷ In this role, any biases introduced by the model are removed—or at least mitigated—by the social scientific practices already in place to address human biases. Thanks to this LLMs can be considered useful tools for finding sources.

However, finding literature is not the only issue the IPCC faces. Using LLMs to flag relevant research leaves reviewers just as overwhelmed as before because the consideration and analysis of evidence is more time consuming than searching for materials. It could be argued that constraining LLMs to finding literature is an underutilization of AI capabilities. A survey of 226 medical and paramedical researchers found that over half believed LLMs would have a major impact on scientific writing (Mishra et al., 2024). The question is, if LLMs could help with reading scientific papers, and writing the IPCC assessments, why not use them? Why constrain LLM usage to only literature search? The next sections explain why expanding the use of LLMs cannot be justified, given their inherent factual inaccuracies and their tendency to erode nuance.

²⁵ Although it is worth mentioning that people do not necessarily use search engines and LLMs the same way. Research has found that people who use LLMs instead of search engines gain shallower knowledge (Melumad & Yun, 2025).

²⁶ One that LLMs may be better than traditional search engines is that their semantic capabilities may allow them to find more relevant sources. Although, again, it is difficult to differentiate between AI and search engine capabilities as these are increasingly intertwined.

²⁷ It may also be valuable for the IPCC to consider other ways to finding relevant sources, such as author reference networks. Thank you to Dr. A. Hamilton-Wright for this remark.

2.3 Why LLMs Should Not Read

I will argue in this section that the IPCC should not use LLMs to summarize information from collected sources.²⁸ In other words, LLMs should not be used by IPCC authors to ‘read’ scientific papers.²⁹ This is because LLMs are inherently prone to factual inaccuracies, also known as hallucinations. But because many developers are working to mitigate the issue of factual inaccuracies, I will argue for two ways scientific language usage makes LLMs more likely to produce factual inaccuracies, even with alternative LLM designs. This is simply to establish that factual inaccuracies are indeed an inherent limitation. First, I will argue LLMs inherently struggle with uncommon words and novel words, both of which occur often in scientific writing. Second, I will argue that factual inaccuracies are more likely because scientists in different disciplines use the same words in different contexts; not only does this mean LLMs perform worse in this context, but it also means LLM usage removes the valuable process of creating zones of agreement on terminology that allows scientific knowledge to be passed between disciplines. After, I will argue for why it is significant for the IPCC that LLMs are factually inaccurate - not just because LLMs *are* wrong, but because it is difficult to ascertain *when* they are wrong. This means relying on summaries from LLMs makes scientists more likely to believe false information.³⁰

LLMs are trained to generate the most statistically probable text based on their training data. They do not explicitly check if the text they produce matches our knowledge of the world. Although methods to reduce hallucinations can help, there will always be limits to what an LLM has been trained on and the information it can access. When an LLM reaches those limits, it produces confabulations. In other words, even when asked questions about topics for which it has no relevant knowledge, the model still

²⁸ Thanks to Dr. A. Hamilton-Wright for the thoughtful comment that there are gradients between having humans read everything and only using AI to read papers. For example, AI models could be used to score sources rather than summarize them, or we might focus on AI that summarizes while preserving the authors own words. These gradients deserve more thought. However, I believe some of the arguments of this section would still apply. Specifically, the arguments which comes later that difficulties identifying factual inaccuracies undermine the trust we can place in these AI models.

²⁹ While the IPCC does not only assess scientific papers the vast majority of the sources are scientific.

³⁰ There is a trade-off here, scientists can gain information from summaries rather than by reading full papers, however some of that information will be incorrect. The underlying assumption for why this is important is that trust in the IPCC depends on their ability to communicate reliable information.

generates an answer—resulting in confabulated content. More simply, LLMs can only use a word they have seen before, and they can only use a word in a context they have seen it in. For example, if an LLM has seen the training text “the sky is blue” before then it will be able to predict this text. However, if it has never seen the phrase and is asked what colour the sky is, it will still produce an answer, but that answer will be based on other training data. Asking a question outside the realm of the LLMs training data could therefore result in the claim that the sky is green. A recent paper from OpenAI claims these hallucinations are caused by fundamental mathematical constraints and cannot be solved by better data or engineering (Kalai et al., 2025). They say “[l]ike students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty” (Kalai et al., 2025).

While this issue is inherent to the current design of LLMs, where models paraphrase based on information they have, there are many designs that aim to address these issues. One such example is Retrieval-Augmented Generation (RAG). RAG allows LLMs to receive and incorporate new information into their training data; new information is typically prioritized over old data (Lewis et al., 2021). This reduces hallucinations as LLMs are able to summarize search results rather than relying on their pretrained ‘knowledge’. For instance, if you were to ask a pretrained LLM about today’s news, it would be forced to fabricate information because it would only “know” what was provided to it in training (i.e. information from large amounts of language from the internet, with the latest entries dated to 2021). However, with RAG, LLMs can access information about today’s news and produce an output based on the new information. The large effort in the AI community focused on solving the issue of factual inaccuracies suggests factual inaccuracies may not always be an inherent limitation of LLMs. However, while it may be possible to remove factual inaccuracies for well-established information - such as the colour of the sky - it will not be possible to remove factual inaccuracies for all scientific papers because of scientific writing practices. In other words, scientific writing is ill suited for LLM summarization.³¹

³¹ This takes on a greater political dimension when considering that industries who want to discredit scientific information will publish malicious literature calling the dominate scientific theories into question (Conway & Oreskes, 2010). AI models therefore have to contend with factual inaccuracies from malicious

The first reason why scientific writing is ill suited for LLM summarization is that LLMs inherently perform worse for less common words. This relates to the issue of bias. Because LLM's "learn" based on the words they are presented with in their training data, if there are fewer instances of a word in the training data, they have less context for how to use that word appropriately. This issue can be "solved" by training LLMs specifically on scientific papers, making scientific terminology more common in the training. However, it is difficult to know if there will be enough instances of a word in a dataset to allow an LLM to use it in the proper context or give the correct information about a word. This is an issue especially for new words (or acronyms), which scientists often introduce. For example, finding a new ant, introduces a novel name, 'Parasyscia ganeshiahi' to the scientific literature (Punnath et al., 2021). At the beginning, an LLM would have no reference for how to use this name other than the singular scientific paper it is defined in.

As new scientific terms have been recently defined, there are not many instances of the word that can be used to train a model. Words that are novel to science will also be novel to LLMs. It has been established that current LLM model performance on translation significantly degrades when translating novel words (Sennrich et al., 2016; Zheng et al., 2024). And they have also been found to less accurately summarize papers with novel words (Balde et al. 2025). This indicates LLMs fail to capture the semantic usage of novel words. Proposed solutions to the problem of novel words, often referred to as "vocabulary adaption," typically assume the word is only novel to the LLM, not to the language, and rely on additional information about the word such as dictionary definitions (Balde et al. 2025). It is likely that additional information is not available when a word is novel to the entire language and not just the LLM. Ultimately, what this demonstrates is architectural constraints of LLMs mean they are less capable of understanding and generating language with novel words than humans. As a result of not capturing the meaning of novel words, LLMs are more likely to create factual inaccuracies, because confabulations happen when these models reach the limits of their input data.³² My argument is simply that because science (including climate science)

actors as well as the factual inaccuracies from their own limitations. This is discussed in more detail in the next section (§2.4). Thanks to Dr. D. Inkpen for this comment.

³² As discussed previously, if there is not relevant information about a topic LLMs continue picking the probable next word, leading to false information.

introduces an increasing number of novel terms, and because current solutions are ineffective in this context, we should expect LLMs to be more likely to introduce factual inaccuracies when summarizing scientific papers.

The second issue for LLMs is that different scientific domains will use the same term in different contexts. There is a substantial literature in science and technology studies that investigates the boundaries of scientific language communities. The historian of science Peter Galison (1999), for example, argues that science is disunified: scientific communities have different concepts and practices, and the shared terminology they use have different definitions. Galison provides the example of the many uses of the term ‘mass’ that have been used in physics as both theoretical and experimental approaches developed. He argues that when scientists from different disciplines work together, they trade scientific objects, where they may, “altogether disagree about the implications of the information exchanged or its epistemic status” (Galison, 1999, p. 146). In this trading process, both sides discuss phenomena together and reach agreements within this localized context. It is not difficult to connect the relevance of this to the IPCC. Climate science includes work from many different sciences and subdisciplines including biology, chemistry, physics, oceanography, environmental science, etc. The IPCC aims to present the findings of these varied disciplines in one report, thus in the IPCC scientific experts must agree upon scientific terms across different disciplines in order to combine scientific knowledge in the form of the final assessment.

This is significant because LLMs perform poorly on words that have multiple meanings, such as the word “ring,” which can mean a piece of jewelry, or a circle, or a noise.³³ If LLMs fail to recognize differences in the meaning of a word based on context, they are more likely to misrepresent information, leading to factual inaccuracies. The task of differentiating between different meanings of a word is called Word Sense Disambiguation (WSD). In general, LLMs struggle with WSD because of the way they are trained; distinctions between meaning may be picked up by surrounding context clues, but LLMs do not explicitly focus on this. The problem can be illustrated by studies showing that, e.g., LLMs perform significantly worse than 16-year-olds in differentiating new meanings of common words, with ChatGPT 3.5 being 22.3% worse (Wu et al.,

³³ Or a set with two mathematical operations (specifically addition and multiplication), or a group.

2024); and have been found to be outperformed by simple clustering approaches (Mosolova et al., 2025).³⁴ This means LLM models are more likely to misrepresent terminology, leading to factual inaccuracies.

Thus far I have focused on establishing two reasons to expect LLM summaries of scientific information to introduce factual inaccuracies: novel words, and words with multiple meanings. However, it is not immediately clear why factual inaccuracies are significant. Even if LLMs are sometimes wrong, in general they seem to perform at fairly high standards – and performance is only likely to improve as development continues. Why shouldn't the IPCC use LLMs to summarize papers, at the very least on a first pass of documents, to help them better understand the state of the literature? One easy rebuttal is that abstracts are already expected in scientific writing, and as such the papers have typically already been summarized by the writers themselves. But, for the sake of argument, let's assume that either abstracts do not exist, or are insufficient to understand a paper. What's wrong with LLM summaries on a first pass?

Factual inaccuracies are a deeper problem because they undermine our ability to know when we can trust the knowledge LLMs provide. LLMs do not provide a guide for when they are wrong, the information is presented the same way regardless of whether it is correct. This means if scientists trust LLM summaries they are more likely to believe false information. This issue is not solved even if LLMs rely only on quotes directly from papers because factual inaccuracies can be produced by omitting relevant information. So, the software used can predictably create ignorance.

There are many examples throughout the history of science of software obscuring information and producing ignorance in ways that, after the fact, are predictable. And we should heed the lessons from these cases. Consider the discovery of the hole in the Ozone layer over Antarctica. In 1985, the British Antarctic Survey announced an area over Antarctica with a severe depletion of ozone. The discovery had actually been made 4 years prior; the team of scientists took 4 years to validate the data because it did not match current scientific hypothesis or more reliable satellite measurements. Further

³⁴ As far as I am aware no recent research focuses on WSD in scientific literature. There is a paper from 2005, which found that models with higher success for WSD in the biomedical domain rely on statistics, and established knowledge in thesauri or dictionaries rather than LLM methods (Schuemie et al., 2005). However, I would be shocked if performance in WSD has not improved since 2005.

investigation of the satellite data revealed that it had also detected the hole in the ozone, but because satellites don't just detect data – they also process it – the low measurements had been rejected as bad data (Conway & Oreskes, 2010). This was not an unreasonable specification for the satellite software; the measurements *were* far outside the range of measurements that could be predicted by scientific theories; translations between analogue and digital measurements require methods to process noise and instrument failures.

There is not a one-to-one comparison that can be made between satellite data processing and LLMs. Rather satellites illustrate how software can work with predictable results to obscure novel findings; recall that LLMs also struggle with novel words. LLMs, and AI in general, thrive on patterns, identifying what has happened previously and using that as the guide for what is acceptable currently. As long as the past matches the present, AI models will be accurate. Novel information is often the most useful or contested information - especially in climate science where new findings such as the Ozone hole may result from recent anthropogenic damage that occurs rapidly and sharply deviates from past patterns - unfortunately this is also where AI models are most likely to fail. LLMs may obscure new information, presenting it inaccurately or not at all, with no indication of when it might be wrong.

In this section I have aimed to argue that LLMs should not be used to read scientific papers. I have argued that LLMs are more likely to misinterpret scientific papers than scientists because they have worse capabilities than humans to understand scientific writing. This is significant because these factual inaccuracies, if believed by scientists, would undermine the reliability of information in the IPCC reports. However, we can also make an argument that goes beyond the accuracy of scientific information. It can also be argued that having LLMs summarize scientific writing overwrites valuable scientific processes that help the production of knowledge.

Peter Gallison (1999) argues the process of agreeing on scientific terms allows for new knowledge to be produced. In agreeing upon terms, scientists produce an agreed upon language, which Galison refers to as a “pidgin”.³⁵ Pidgins, Galison argues, allow

³⁵ A term from linguistics that refers to a simplified language that develops between groups without common language for basic communication such as trade.

scientific disciplines to trade knowledge. For examples, Gallison focuses on differences between theoretical and experimental physics, specifically in the context of WWII. During this period, laboratory work brought together theoretical and experimental physicists, who collaborated closely. This collaboration acted as a “trading zone,” where ideas from field theory and radio engineering were exchanged, ultimately leading to the development of radiology as a new discipline. Scientific communities may not agree about terms and information, or their epistemic value, however when they engage nonetheless the valuable process of knowledge transfer can occur. In actually having to write summary reports, humans from scientific communities with different language have to work with one another to find commonalities. This process itself is significant. This means if the IPCC uses LLMs to read scientific reports not only will LLMs sometimes be factually incorrect, but their use will also obscure the valuable processes of agreement across scientific disciplines that allows knowledge to be shared. This line of argument, that LLMs should not be used (1) because they are ill equipped for scientific communication *and* (2) because they overwrite valuable scientific processes will be continued in the next section.

2.4 Why LLMs Should Not Write

This section argues that the IPCC should not rely on LLMs to write scientific assessments because they erode the communication of calibrated uncertainty and the conditions of trustworthiness, which are central to the IPCC’s goal of communicating climate science to policy makers. My argument will run as follows: It is difficult to turn scientific findings into policy-relevant communication in part because science and policy demand different styles of thinking and writing. Scientists aim (or are trained) to recognize uncertainty and disagreement, but policy demands (and the public expects) some kind of consensus about what we know. When it comes to assessments, resolving this difficulty means being explicit and careful about: (1) what words are chosen (e.g., being extremely precise about the communication of uncertainty in the produced report), and (2) how those words are arrived at (e.g., having the produced report be a result of a deliberative process that legitimizes its conclusions). I argue that LLMs do not fare well in either endeavor: (on 1) they lack the precision required; (on 2) they forego the important

process of deliberative acceptance.³⁶ Having sketched the case against LLMs, let's now turn to the first piece: scientific writing's commitment to recognizing uncertainty and disagreement.

Scientific writing aims to recognize uncertainty and disagreement. Uncertainty is important to scientific writing because although it is common to talk about “scientific facts” as beyond any doubt, this is better phrased as “beyond any *reasonable* doubt” because the conclusions of empirical science, any science whose conclusions are based on inductive reasoning, will always remain to some degree uncertain. This is true, e.g., for all sciences employing statistical testing.³⁷ Scientific language is also careful to communicate disagreements. This is because disagreement is important to the production of scientific knowledge; it helps scientists ensure rigor, identify flaws, prevent dogmatism, and maintain trust through transparency. Consider the ozone case from the previous section. Scientists noticed disagreement between weather balloon measurements and satellite data. Pursuing that discrepancy, they uncovered the ozone hole, which existing theories could not account for.

While uncertainty and disagreement are important within science, in policy, uncertainty and disagreement can undermine public confidence in both experts and science. Policy makers are often asking for information that will help guide them to a decision – they expect some form of consensus about what will happen if they make a certain decision. For example, they may expect the IPCC to provide a clear indication of how much carbon dioxide can be safely released into the atmosphere, and what the impacts will be if we exceed a safe emission limit. In their book *Discerning Experts*, Oppenheimer et al. (2019) explore the mismatch between scientific and policy

³⁶ An additional concern: one strength of experts is that they can tailor their delivery for specific audiences. Such that an expert can effectively communicate their ideas to a child, a politician, or another expert. To do so the expert must gauge how much knowledge and capacity for understanding their audience has and communicate accordingly. LLMs do not currently have a way to take on different voices for different purposes and thus may not live up to the task of scientific communication in all environments. This is not the focus of this chapter, as scientific assessments aim to write reports for policymakers, so the audience is well defined. But it is important to consider when using LLMs for scientific communication. Thank you to Dr. A. Hamilton-Wright for bringing it up.

³⁷ Scientific work which collects data and then uses statistical tests (resulting in p-values) to determine the results of their findings has to be particularly careful to clearly communicate the certainty of results based on these statistical tests. In part because people are notoriously bad at understanding probabilistic reasoning.

communications. In the realm of policy, Oppenheimer et al. say that “[c]onsensus appears to provide a way of signaling the agreement of experts about what knowledge is important enough and sufficiently settled to inform policy making. [...] It allows scientists – or at least discrete groups of scientists – to speak with a collective voice. The unstated suggestion is that if this voice is univocal, then the science must be reasonably secure. Conversely, if scientists cannot reach consensus, this might be viewed as a signal that the science is insufficiently settled to inform public policy” (Oppenheimer et al., 2019, p. 11). This makes scientific assessments tricky, as they are trying to balance conflicting goals in science and policy.

Science communication is tricky, but in many cases it’s critical. Getting scientific communication wrong can mean losing public trust in scientific information, opening policy decisions to malicious exploitation. Oil and gas companies have undermined climate science by over-emphasizing uncertainty and disagreement in a variety of ways over the years. These methods – including funding “contrarian” science and think tanks, creating front groups with scientific sounding names, and running marketing campaigns – are detailed in the book *Merchants of Doubt* (2010), by Conway and Oreskes. For example, Conway and Oreskes document how the George C. Marshall Institute issued reports in the late 1980s attacking climate science as it gained public attention. The institute circulated unpublished “white papers” claiming that solar activity, not human actions, caused global warming. It also presented similar arguments to the White House. These efforts sowed doubt about anthropogenic climate change and helped delay policies to reduce emissions.

For the IPCC, the solution to issues raised by overemphasis of scientific uncertainty and disagreement is to clearly communicate exactly what is (and is not) uncertain through a strict and clear procedure. This is also relevant to the IPCC’s aim to inform policy, not dictate it. The IPCC does not want to make political arguments; it aims to provide policy makers with information that will help them make decisions.³⁸ As Betz (2013) argues, by having precise procedures for communicating uncertainty the IPCC can

³⁸ The boundary between science and policy is penumbral. The IPCC’s approach evolves from approaches established by ozone assessments (Oppenheimer et al., 2019). If science should aim to be policy neutral, or if it can even be policy neutral are separate debates. Or, if not separate, unrelated enough that I believe I can avoid discussing them, especially since the IPCC has a fairly clear stance on this.

communicate the current state of understanding without unduly imposing its own set of values. This means the IPCC can communicate potential consequences of policy decisions without explicitly taking a political stance on the matter; even when scientists, “cannot reliably forecast the effects of the available policy options [...] in a probabilistic—let alone deterministic —way.” (Betz, 2013, p. 215-216). For example, allowing scientists to provide information about safe levels of carbon emission despite models that disagree about the precise impact of those emissions.

The IPCC's procedures for communicating uncertainty emphasize two key factors: the amount of available evidence and scientists' confidence in that evidence. This process, illustrated in Figure 4, requires authors to make explicit judgments at each step. They first assess whether sufficient evidence and agreement exist to evaluate a claim's confidence. Next, they identify claims with consensus even amid limited evidence. Finally, with robust evidence, they gauge confidence levels and the likelihood of specific outcomes. Importantly, because scientists are involved in making explicit judgements the process of deliberative acceptance (this term will be discussed in more detail later) is possible. The value of scientific agreement still rests on scientific disagreement. While scientific disagreements are not explicitly stated, they are the underlying mechanism that legitimizes the final product. Scientists in the IPCC are not just communicating the evidence; they are communicating the opinions the scientific community has about that evidence.

The question is whether LLMs can be used to write the IPCC reports. This can be divided into two separate questions (1) can an LLM use the IPCC procedure to communicate the level of uncertainty, and (2) can an LLM determine the opinions a scientific community has about evidence.

I argue that an LLM cannot use the IPCC procedure to correctly communicate the level of uncertainty. LLMs pick words with a stochastic probabilistic process. For any given input sequence, the model produces a probability distribution for every word in its vocabulary, with higher likelihoods for words that commonly follow based on training patterns.

Evaluation and communication of degree of certainty in AR6 findings

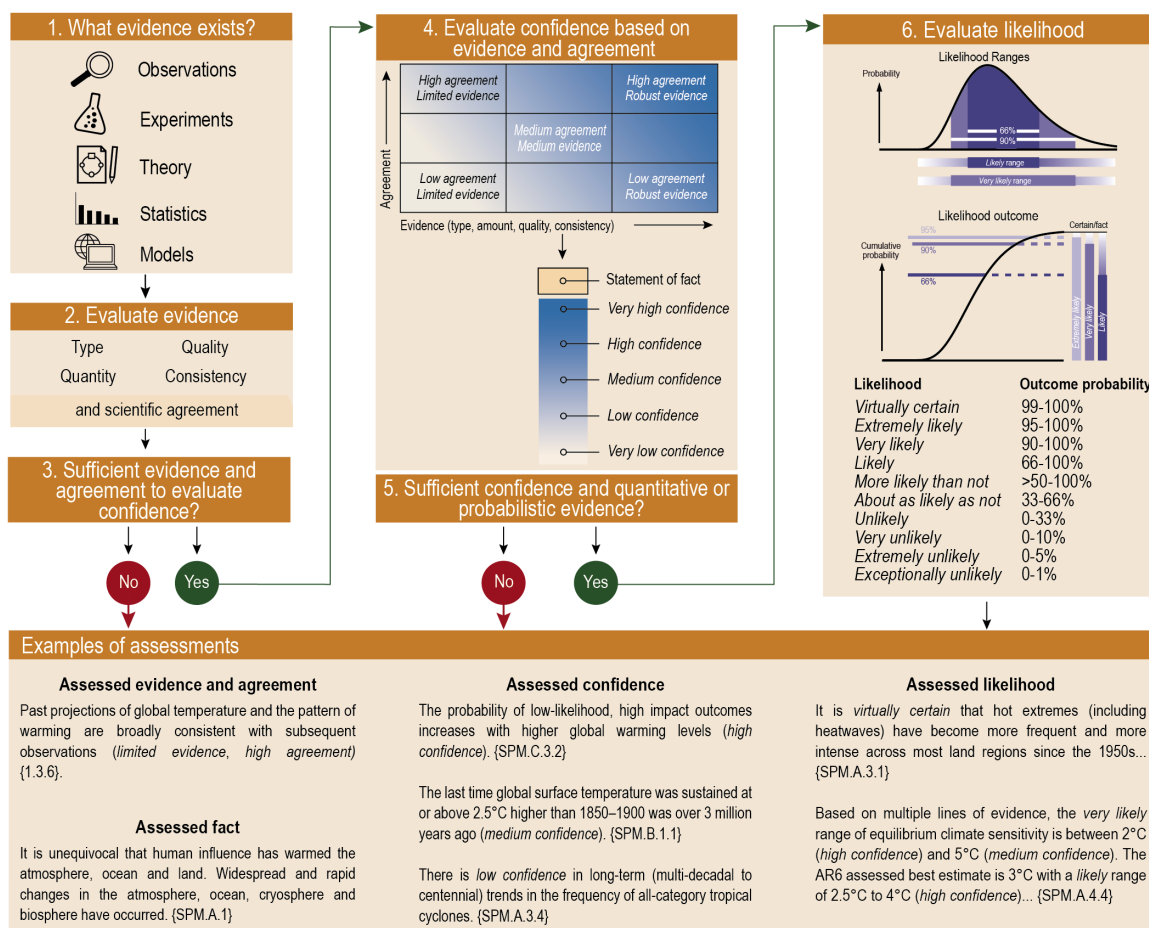


Figure 4. IPCC Diagram for the Evaluation and Communication of Degree of Certainty in AR6 Findings (IPCC, 2021)

It then samples a word from this distribution using randomized methods like top-k or nucleus sampling, rather than always picking the most probable one. LLMs use this approach to produce human-like text instead of repetitive outputs from deterministic selection (e.g., greedy search). This means that LLMs choose words in ways that are randomly determined. This produces a challenge for the IPCC's process because the wording should be consistently chosen.

Because the language of the IPCC is so precisely calibrated, changing the wording slightly changes the meaning significantly. The IPCC uses carefully defined terms like “very likely” to mean a 90-100% probability, “likely” for 66-100%, often these terms are italicized in the reports to indicate assessed likelihoods (IPCC, 2021). Changing

very likely to just *likely*, or omitting these qualifiers altogether, would alter the communicated certainty and risk assessment substantially. For example, the sixth assessment’s summary for policy makers states that, “Human influence is *very likely* the main driver of the global retreat of glaciers since the 1990s and the decrease in Arctic Sea ice area between 1979–1988 and 2010–2019” (IPCC, 2023, p. 5). If an LLM had written *likely* instead of *very likely* it would convey a very different level of certainty about the claim. Even if an LLM can assess what the right option should be, there is still a chance that it will not select this option, because of its stochastic nature. This means there is no guarantee an LLM picks the correct word according to the IPCC procedure.³⁹

It is possible to imagine solutions to this issue.⁴⁰ Instead of stochastically sampling from the probability distribution the LLM could pick the most likely word. Although, this would remove the ability of the LLM to produce human-like language, making it less useful for writing the rest of the report. To solve this, we could further imagine that the LLM would only pick the most likely word when the word is one specified by the IPCC procedure. This task would likely not be trivial to solve but may be possible. While no models⁴¹ switch between stochastic and deterministic methods based on preset words or context, many dynamically modulate stochasticity. However, it is not only important that LLMs can follow the IPCC procedure consistently. It is also important that the wording the LLM selects reflects the perspective of the scientific community. This brings us to the second question: can an LLM determine the opinions a scientific community has about evidence?

I argue that LLMs cannot determine the opinions of a scientific community. While LLMs may have access to the scientific papers under assessment, they lack direct knowledge of the scientific community’s consensus or stances on that evidence. A study by Algaba et al. (2025) found that LLMs tend to favor papers that are highly cited, have shorter titles, and fewer authors. This suggests that LLM-generated reports for

³⁹ This is also an issue for using LLMs to summarize the IPCC reports – which people may wish to do as the length of the reports have grown over time. Stochasticity means the LLM may misrepresent terminology used to communicate uncertainty, changing the meaning significantly.

⁴⁰ Although interestingly newer LLM models are more likely to generalize scientific findings than older models (Peters & Chin-Yee, 2025). Suggesting that solving this issue is not a focus of many developers.

⁴¹ Or at least none I am aware of.

organizations like the IPCC would disproportionately represent well-cited papers.⁴² Additionally, while LLMs are currently effective at identifying language errors in scientific papers, they struggle to distinguish between when scientists are using careful language because of scientific writing practices and when their language is because of low confidence in their results (Lee et al., 2025). Taken together this indicates that LLMs struggle to assess the legitimacy of scientific findings and rely on metrics such as citation numbers to determine credibility. This indicates LLMs are likely to represent knowledge from unreliable sources (such as malicious grey literature) as ideas that are scientifically valid, especially if these sources are well cited.⁴³ This is important because, as noted by Conway and Oreskes (2010), malicious grey literature has historically been used to create uncertainty around scientific findings. If AI models cannot distinguish between reliable and unreliable scientific information, we ought to expect it to easily fall prey to the industry denialist tactic of publishing disinformation.

For the sake of argument, let us assume that LLMs can determine the reliability of scientific information. Perhaps they simply get better at assessing the reliability of evidence thanks to new advancements; they could rely on better metrics such as journal impact factor;⁴⁴ or scientists could individually provide their own written assessments of the evidence, which LLMs synthesize into consensus statements. In this case, LLMs still should not write scientific reports because agreement is not the only goal of scientific assessments. As argued by Beaty and Moore (2010) it is also important how scientific agreement is reached. They argue scientific knowledge is legitimized by the process of deliberative acceptance, which involves, spaces for disagreement, deliberative equality, and majority rule. In science, scientific knowledge is valid when all individuals involved were heard and that all agreed that, regardless of whether the outcome went the way they wanted, they would stand by it.

⁴² This type of bias is bad for science. Papers that cover more complicated and/or technical subjects tend to be longer. Works that do less collaborative research will have less authors. Thank you to Dr. D Inkpen for this comment.

⁴³ As far as I could find, no research has specifically examined LLM preferences for grey literature, such as preprints and technical reports

⁴⁴ Although these have increasingly been shown to be prone to manipulation and bias (Singh Chawla, 2024).

Producing consensus statements with LLMs removes deliberative acceptance from the IPCC reports.⁴⁵ AI consensus seems to forego this process, because there is no method under which agreement is reached, the LLM simply makes a final statement. If LLMs were used to write IPCC reports there would no longer be proper deliberation, instead an LLM would opaquely output levels of certainty; conversations and disagreements would be obscured by the process of LLM arbitration. But these disagreements are important to the production of knowledge and its trustworthiness.

While some scientific disagreements will be epistemic in nature such as disagreement about scientific terms or methodology between disciplines – others disagreements will be value-based. Value based here meaning non-epistemic values such as social, ethical, or political values, that are not related to truth, but instead to things like what is just, right, good, etc. This is the meaning people typically refer to when using the term “values”. However, technically both epistemic and non-epistemic decisions are “value-based” when understanding “values” as the basis on which we make decisions.

Some IPCC disagreements will be value-based because the IPCC aims to describe a wide range of information needed to understand climate change (IPCC, 2013a). How these topics are reported and represented depends on what we value, for example economic stability or environmental stability, or the current standards of living as compared to the standards of living for people in the future. The framing of scientific information thus has the ability to shift political decision making in different directions based on underlying non-epistemic values. As Longino (1990) argues, the weighing of values is why diversity has epistemic merit in science. A diverse set of people will have different perspectives and values. This in conjunction with the process of deliberative acceptance leads to greater objectivity (where knowledge is less representative of any particular person’s point of view).

⁴⁵ One solution to this may be for AI models to engage in deliberative acceptance. However, as Dr. R. Moser thoughtfully brought up (personal communication) it is unclear what it would mean for an AI model to disagree with the outcome of a decision but stand by it because of the process involved. This would require conceptualizing AI models, as separate entities that can hold positions about scientific knowledge. Perhaps we could train sets of AI models to hold different stances and engage in deliberative acceptance, but this would be quite a departure from the way LLMs are used and conceptualized today, and it is not clear this would be similar to, or the same as, the human processes of deliberative acceptance.

It is not clear how AI models address diverse values, further it is not clear they have access to a diverse set of values. For example, Indigenous knowledge is centralized in knowledge holders and transferred orally – it is not clear how an LLM would even access this knowledge, much less evaluate it.⁴⁶ This is significant as the transparency of disagreements is what legitimizes scientific knowledge. That scientists have room to disagree and be heard while producing reports is what makes the end results valid information, even if that disagreement is not represented in the final report – the process legitimizes the outcome. Why then should we consider the consensus statements LLMs can produce to be reliable, trustworthy, or legitimate? If we wanted to accept AI generated consensus statements, there needs to be some explanation of how a model can and should replace the process of scientific agreement, and currently no such argument exists.⁴⁷

In summary, LLMs simply do not have the precision of language, or recognition of human values that are necessary for the writing and communication of IPCC reports. For this reason, they should not be used to write the IPCC reports. However, this does not mean they should not aid the writing process at all. LLMs may be helpful tools to help edit IPCC reports, as long as authors are careful to preserve their meaning in the editing process. Some evidence suggests that LLMs may be especially helpful for improving clarity in writing produced by non-native English speakers (Lee et al., 2025).

2.5 Conclusion

In this chapter I have argued for how the IPCC should use LLMs; specifically, I have argued that while LLMs can be useful in finding literature, they should not be used to read or write IPCC reports. My arguments about where LLMs should be used can also be understood as an argument for minimal automation.⁴⁸ That is, while LLMs can be viewed

⁴⁶ Thank you to Dr. D. Inkpen for articulating this so clearly in our discussions.

⁴⁷ This conversation changes if you consider AI as an agent capable of reasoning and engaging in the scientific process (which LLMs do not explicitly aim to do). People are working towards artificial general intelligence (AGI) which can reason like humans, but it is generally recognized that we are not there yet. Even if we were there, you would have to prove (somehow) that an AGI agent would be better than a scientific community.

⁴⁸ This would be more in-line with arguments Al Khourdajie (2025) makes that the integration of AI should recognize the difference between two different institutional roles that the IPCC plays: the role of a ‘producer’ and ‘assessor’. For the first role, the IPCC as a producer, while AI systems would be integrated

as tools to speed up or improve some portions of scientific assessment (such as finding literature, or editing), they should not be considered effective ways to automate portions of scientific assessments that require human input. I have argued this on two main bases. The first is that LLM language capabilities are mismatched with the practices of scientific writing. The second is that high automation in IPCC reports erases valuable human processes in knowledge production (such as deliberative acceptance, or the production of ‘trading zones’). This argument has not been made on the basis that there is some human capability that makes us uniquely capable of completing scientific processes.⁴⁹ Rather it rests on the idea that we trust scientific information because of factors that are inherently social, and we have not considered in enough depth the way LLMs fit into these complex social assessments of trust to justify their use.

I have aimed to argue my position not from the current state of LLM capability, but rather from a hypothetical ‘best case’ where LLM capabilities have improved beyond what is currently possible.⁵⁰ While I think this is important, it is worth noting that many of the arguments that AI should do more in decision making capacities depend on a belief that AI models have capabilities that they do not possess, or the argument that they could one day possess those capabilities. These arguments are insufficient for current adoption of AI for higher automation purposes in scientific assessments. To use LLMs in the contexts I have argued against (reading and writing), one would have to demonstrate LLMs are better than humans, and that human processes are not more important than the performance increase LLMs may provide. This would have to be done practically, not hypothetically.

In discussing the IPCCs adoption of AI I would be remiss to not mention the conflicting values of doing so. While the IPCC focuses on highlighting the dangers and realities of climate change, AI systems themselves can produce high levels of carbon

into the IPCC workflow, the emphasis would remain on humans to utilize these tools to develop reports. Alternatively, the second role the IPCC plays—the IPCC as an assessor—involves the IPCC using AI models to produce and/or summarize portions of these reports. I have essentially argued that AI is okay in the role of the ‘producer’ but unacceptable in the role of the ‘assessor’.

⁴⁹ Discussion about what skills qualifies a machine for personhood, or rather what distinguishes human capabilities from AI ones are quite complex and would open another avenue in the already wide range of literature being considered in this thesis.

⁵⁰ This was done to be able to think about the application of these models at their best, as opposed to focusing on a taxonomy of the ways that these models might need to improve. Essentially asking, if we believe that these models will continue improving, what is the limit to their application?

emissions (Xiao et al, 2025). It would be strange if an organization aimed at informing policy makers about the dangers of climate change appeared to endorse an industry contributing to the problem.

In general, it is worth noting that integration of AI in the IPCC will depend on public perception and trust of AI systems. Buck et al. (2025) argue that implementing AI to a level that does not match public trust will undermine the legitimacy of the IPCC. For example, they hypothesize a scenario where there is little trust in AI and AI is used to automate large portions of the IPCC – resulting in people not trusting the outcomes of the IPCC assessment. Alternatively, they argue if there is high trust in AI and the IPCC does not utilize it then the IPCC could be critiqued when their produced report does not match reports produced by AI models, leading to doubt in the IPCCs scientific authority. This is important because the IPCCs effectiveness relies on policy makers trusting the scientific report enough to use it to inform their decisions. Chapter 3 will further discuss if we can justify our trust in AI models by examining the underlying assumptions we must accept to trust these models.

3. Conclusion: A Challenge to the AI Paradigm

As artificial intelligence (AI) models are able to do more and more, some seem inclined to treat them like experts. Previous chapters of this thesis have argued that AI models and scientists cannot fulfill the same roles in our decision making. Chapter 1 demonstrated that methodological similarities between reasoning in medical science and the generation of AI outputs are not enough to justify the usage of AI. Chapter 2 further established that AI models diverge from scientific methodology because they focus on knowledge output rather than the process of its production. This chapter aims to demonstrate, first, that assumptions underlying AI reliability depend on human values, but second, and unlike science, the production of AI models does not even attempt to address these values in clear, transparent, or communal ways. I will argue that we should be careful with our usage of AI models because of the problem of inductive risk – which argues that values are required in science when the non-epistemic consequences of error need to be considered.

The argument will proceed as follows. I will first aim to establish that the production of AI information, like science, must involve human values. I will do this by examining two assumptions we make when using data to make predictions: (1) we can choose data that will accurately represent the world, and (2) the world follows stable patterns that will persist. For each assumption I will address why it relies on human value assessments. I will then compare the AI case to science, arguing that while science has methodologies that address human values, the production of AI models, and by extension the way those models generate output, do not. The consequence of these arguments is that we ought to treat AI information differently than scientific knowledge.

When we can verify that our assumptions about data and patterns are accurate, AI is well-suited for decision making. However, in many scientific decision-making contexts, these assumptions fail—often because the goals involve value-based decisions with complex trade-offs. While this unpredictability affects human and AI systems alike, established scientific and social frameworks already possess mechanisms to navigate these decisions, whereas the employment of AI systems does not.

3.1 Challenging Assumptions of AI Predictions

This section will challenge the assumptions that: (1) we can choose data that will accurately represent the world; (2) AI can identify stable patterns in the world that will persist. For each of these assumptions, it is possible to treat them as an ontological claim – e.g., that the world is able to be represented as data, or the world follows stable patterns. I will aim to assess them from an epistemic standpoint. Arguing that regardless of the underlying state of the world, our ability to assess the accuracy of AI information is contingent on our knowledge and perspectives.

3.1.1 *Questioning if we Can Represent the World in Data*

For AI predictions to be reliable it must be true that data can represent the world.⁵¹ We must believe that collected data in some way corresponds to items in the physical world. For example, we may test whether a measurement of 5 foot 6 inches accurately represents a person's height, we believe we can select data that will lead to reliable predictions because we believe data can be representative of the truth.⁵² This is most explicit in supervised models, where labeled data acts as the ground truth. But even in unsupervised models, we assume AI outputs are valuable because they find patterns in the data that are present in the world. However, the data we provide AI models with will always represent “our” interpretation of the world. This is because we have to make decisions about what data to exclude and include – for practical reasons and epistemic ones. In other words, the data we give AI will always contain value decisions made by people.

There are practical reasons why we can only provide a limited amount of data. Limitations in memory and funding mean we cannot provide all the data in the world to AI models. Because of the cost of memory storage, we cannot collect data on everything, and because of training costs we cannot always provide AI models with all the data we can collect. As we provide larger amounts of data to AI models, these models get more expensive. More data requires more computational power, energy, and time to process

⁵¹ Or at least represent the world enough to lead to accurate predictions.

⁵² I recognize it is difficult to define what it means for something to be “true”, as well as how we might establish that something is “true”. This word is used here intentionally because we are assuming data can represent “the truth” despite issues with what this might mean.

information. For many smaller companies this means they have to make decisions about which data gets used. Especially because collecting clean data is a costly part of the AI development process. A study of CIOs by Salesforce found that companies integrating AI typically spend 20% of their budget on data infrastructure and management, and only 5% on AI development (Salesforce, 2024).

However, many AI companies developing LLMs are receiving unprecedented amounts of funding. Between 2013 and 2024, total global corporate investment in AI reached \$1.6 trillion (Stanford HAI, 2026), and is forecasted to rise to \$2.52 trillion in 2026 (Gartner, 2026).⁵³ This makes financial constraints seem trivial – these companies can afford to collect and train on huge amounts of data. The size of datasets used to train large language models (LLMs) has more than doubled every year, with models today training on datasets of tens of trillions of words (Rahman, 2026).

Yet even with well-funded AI companies, value decisions about data are made. For example, most datasets for LLMs focus on mainstream American English (Louro, 2025). Further these datasets come from the public internet. A choice that is made largely because of the ease of collection. Internet language is (arguably) free to use, and already digital. There is no need to transcribe spoken language or compensate people whose language is used.⁵⁴ Both of these decisions are value decisions, which exclude people who do not speak like online American English speakers. American English is not representative of all English speakers, and public online text represents a particular type of speech – one that has different conventions from casual spoken conversations (McCulloch, 2020).

As the size of LLM datasets have ballooned, some have questioned if AI companies will exhaust the language that is available online (Educating Silicon, 2024; Villalobos, 2024). Villalobos et al. (2024) estimates that companies may use all the available public English language data by 2036. An accurate estimate of the amount of language available online, including language that has not yet been posted is incredibly

⁵³ For comparison: the Manhattan project - which developed the first atomic bomb – cost roughly \$36 billion (adjusted for inflation); building the entire interstate highway system in the USA costs roughly \$620 billion (Hussein, 2026).

⁵⁴ Although whether people deserve compensation has more to do with the status quo than ethical or moral obligations.

difficult to determine, so these claims are largely speculation. However, if companies do exhaust the language available online, they will have to decide how/if to collect more data. This may include using synthesized language,⁵⁵ private online messages,⁵⁶ or collecting spoken language that is not documented online. Making any of these decisions is value laden, and it will lead to the LLM reflecting a type of language situated in a particular worldview.

Beyond the practical reasons of memory, funding, and data availability. There are also epistemic reasons that collected data will always be value laden. This is because the world is analog (or continuous) but computers store information digitally (discretely).⁵⁷ To convert from analog to digital we must make, at some level, decisions about how to convert the analog world to the digital one.

One example of this is the coastline paradox which states that there is no well-defined length of a coastline. When measuring a coastline, we must decide how fine-tuned a measure to use. Do you measure a straight line along the coast? Or do you follow the jaggedness of the land? Landmasses have features at all scales, an outcropping of rock could be hundreds of kilometers, a few centimeters, or even smaller. There is no obvious size of feature that should be included or excluded from measurement, and therefore no single well-defined perimeter. If you measure a coastline with a meterstick it will be shorter than if you measure it with a ruler.

This paradox is challenged by the fact that we do have measurements for coastlines. For example, Canada's coastline is 71,261 km long (McCann 2026). Further, coastline measurements are made easier by modern technology such as measurements from satellites. However, the coastline paradox does not suggest we cannot arrive at an agreed upon measurement of a coastline. It indicates that even something as simple as measuring a coastline involves a value decision about how much precision we should use. The concept of a coastline is a human construct that depends on a number of agreed upon conventions such as the granularity of measurement, and tidal datums. The coastline of

⁵⁵Though models trained only on their own output have been shown to fail, a phenomenon referred to as model collapse (Gomstyn, 2026).

⁵⁶There was some incorrect reporting that companies like Google and Facebook were going to use private messages to help train their AI models, but this has not happened (Jancer, 2026), or at least not happened yet.

⁵⁷Excluding analogue computers which are less common than their digital counterparts.

Canda is reported to be 243,797 km if you include islands, with 3,800 km of coastline along the Great Lakes (McCann, 2026). Should these additional coastline measurements be reported as part of Canada's overall coastline? This is a decision we must agree upon to determine the length Canada's coastline. Thus, we can see that to measure a coastline we must agree upon measurement conventions, and also what areas can be included as coastline.⁵⁸

The coastline paradox illustrates why we may believe converting the analog world to a digital one requires value judgements. However, this perspective can be challenged if a person does not believe that a conversion from analog to digital is necessary. We may believe we can represent the world in data for metaphysical reasons. For example, the Mathematical Universe Hypothesis (MUH), proposed by Max Tegmark, suggests that our physical reality is not just described by mathematics, but is a mathematical structure (Tegmark, 2008). If all things are mathematical in nature, it follows that we could represent them as data, or functions, potentially in computers.⁵⁹ But, even if we accept this metaphysics, it does not provide a way to determine what data leads to the true mathematical structures of the world. So, even under an ideal metaphysics, we still have to contend with the contingent nature of the data we use to train AI.

To select data that accurately represents the world we must decide which world we are portraying. These decisions will influence the outputs of AI models because AI models depend on their underlying data to act as the representation of the world. But what about when we have agreed upon data, as is the case with measurements of the coastline. Or further, the argument that despite bad data, AI models are still able to identify patterns in the world because these models go beyond the data and "learn" patterns from it. Even then, our trust in AI predictions depends on another assumption: that patterns in the data will persist. The next section addresses this assumption.

⁵⁸ The assessment of these values is connected to the specific application of an assessment, e.g. you would need more precision on the coastline for a rescue operation than for a middle school educational setting. Our interests and goals play a role in how we resolve debates like the coastline paradox. Thank you to Dr. A. Hamilton-Wright for this comment.

⁵⁹ You may also believe that we can represent the world in computers if you are a physicalist that believes the world is comprised of discrete modellable entities.

3.1.2 Questioning if the World Follows Stable Patterns that Will Persist

To trust AI predictions, we must think that the world underneath them is predictable. This means we must think the world is stable, such that predicted patterns persist. Science plays a large role in our conceptualization of the world as stable and predictable. This is especially clear in physics, where theories have been so successful it seems possible the world is ontologically predictable with the use of natural laws. However, even if the world follows patterns, we do not know what all of those patterns are. Therefore, even if the world is ontologically predictable, there is still an epistemic responsibility to prove that predictions are correct. This section aims to establish there are weak patterns, patterns we observe that do not persist, and patterns we observe that we do not want to persist. This demonstrates that our central assumption is not always true: the world does not always follow stable patterns that persist.

First let us examine weak patterns. Sometimes data demonstrates weak patterns, where noise or large variance means many datapoints do not ascribe cleanly to a pattern. This is recognized in biological field studies, which may accept lower p-values in statistical testing because of noisy data with high variances (Fay, 2018). In data with weak patterns, we ought to expect predictions made for an individual to be less accurate or inaccurate. For example, in general it is a trend that men are physically stronger than women, but this does not mean that all men are stronger than all women. Even if a statistical trend exists, we cannot apply it to all individual circumstances. Weak patterns suggest phenomena we cannot predict well based on data. This is a concern for AI applications that try to apply predictions to specific individuals. For example, medical AI models that aim to produce a diagnosis for a specific patient. When the patterns are strong, as is the case with mammography on less dense breast tissue (Mansour et al., 2022), these AI models can be expected to perform reliably. But when the patterns are weaker – as they are with dense breast tissue – we ought to expect worse predictions.

Even when we discover patterns strong enough to support accurate prediction, we must also assume those patterns will persist in order to justify believing that AI predictions are useful. However, there are instances where we know that patterns in the world have not persisted. Consider the rapid deterioration in ozone levels found by climate scientists. CFC emissions into the atmosphere deteriorated ozone in the

atmosphere so quickly that ozone measurements registered as errors in the measurement instruments. The levels were outside the range we thought was possible. This shows a sharp divergence from a trend that had otherwise been stable for a long time. It may be argued that we could predict these changes in patterns with sufficient information. Scientists knew about the effects of CFCs prior to the discovery of the ozone hole. However, when the hole was discovered, they did not have reason to expect the extent of depletion that had occurred. If we claim that we can predict changes, there must be some justification for our ability to do so. Often this justification comes from the evaluation of a variety of information, and value judgements about what they will produce. Trust in the persistence of patterns in the world is a value decision.

This value decision is relevant when considering the application of AI models. Consider language. We expect language to change over time. We might expect the creation of new terms, such as “delulu” – an internet term roughly meaning delusional. We can also expect existing words to change in meaning. For example, the term ‘performative’ has been shifting from referring to the performance of an action, to meaning that an action is performed insincerely.⁶⁰ In chapter 2 we saw that LLMs perform worse for novel words, and distinguishing words with multiple meanings. These are instances where the stability of patterns is weaker. However, this does not suggest that LLMs are not suited for language capabilities overall. Models can be updated with new information, making the instability of language patterns insignificant. Note that deciding shifting patterns in language are not significant enough for LLMs to be ineffective is a value decision. People must assess the performance of the model and determine what counts as acceptable and unacceptable performance. This assessment will be different depending on who is making it.

That people must make value decisions about the strength of patterns when using AI is especially clear with patterns that we do not want to persist. For example, patterns of oppression. It is common for AI models to produce racist and sexist outputs. So many of our sociocultural systems produce racist and sexist results that if AI models identify patterns, these are often the patterns that are found. AI models can even find these

⁶⁰Dr. A. Inkpen comments that a favorite of philosophers is “begs the question” which used to mean “assume what you are trying to prove” but now often means “raises the question”

patterns using data we do not think reveals sensitive information, such as determining race from zip codes – these are rereferred to as proxy features (Goyal et al., 2024). If we use sexist, or racist predictions to make our decisions, those decisions will uphold patterns that we wish to stop.

There may be other patterns we do not want to continue. Like the asthma example from chapter 1, which predicted patients with asthma were less likely to have negative health outcomes from pneumonia. This is true, but it is true because people with asthma are more likely to take the risk of pneumonia more seriously. We do not want people without asthma to continue to underestimate the risk of pneumonia and have more severe health outcomes.

This section (§3.1.2) has aimed to demonstrate that our acceptance of AI information depends on our value judgements about the strength and importance of patterns in the world. There are decisions that need to be made about the strength of a pattern, if we think it will persist, and if we want it to persist. The previous section (§3.1.1) discussed that there are also value decisions we must make when selecting data for AI models to be trained on.

However, it is not clear how AI development and use should assess these value decisions. How should we determine when data is sufficient? Or when high accuracy in AI models correspond to patterns that are useful to know⁶¹ as opposed to ones we wish to stop? As will be discussed in the next section, this is not an issue that is unique to AI, but it is one that is poorly addressed in AI development.

3.2 How is this Different from Science?

Science also faces the challenges of how to contend with human values. Heather Douglas (2000) phrases this as the argument from inductive risk. Douglas claims scientists must make a number of choices in their work, such as choosing methodological approaches, or how to characterize data. For example, when climate scientists assess temperature records, they must decide how to handle urban heat islands – where urban areas produce higher temperature measurements (Cheval et al, 2024). Scientists may decide to apply a

⁶¹ Such as when imaging characteristics of a tumor indicate it is cancerous.

statistical model to “adjust” the temperature readings in cities to make them comparable to rural areas, or they may choose to focus only on rural areas. Choosing between a more comprehensive dataset and a cleaner one.

However, any choice scientists make can be wrong (that is, might turn out not to have been appropriate or correct). This stems from the problem of induction, which roughly claims that because inductive reasoning only follows with probability, rather than certainty, we can never know from our observations that any given conclusion will follow.⁶² Douglas further claims that when science is used to inform public policy, mistakes of scientists have significant consequences. Because of this, scientists must evaluate the consequences of error in their scientific practice. This evaluation requires using non-epistemic (i.e., social, political, or ethical) values. According to Douglas this means that science needs non-epistemic values to assess what the correct decisions are, and to assess risk. Not only is value-free science unattainable, it is also undesirable. Values are in fact required to do science that informs policy. In her work, Douglas explains the various ways in which these contextual values can play a role in science without undermining scientific objectivity, as discussed below.

The problem of inductive risk applies to AI in a similar way. As discussed in the previous section (§3.1), the collection and assessment of data and patterns require value judgements. These judgements can be wrong, and when they are applied to decision-making contexts - like medicine or climate science - those mistakes have consequences. So, AI developers and users⁶³ should evaluate the consequence of errors using contextual values to determine what the correct decisions are.⁶⁴

⁶² In philosophy there is longstanding debate and discussion about the justification of inductive inference (Henderson, 2024).

⁶³ Different levels of knowledge and differing abilities in the power to change AI models mean developers and users will not have the same responsibilities when it comes to assessing AI. However, the specific roles of these different groups are not explicitly explored in this thesis. Al Khourdajie (2025) suggests developers are responsible for addressable limitations, while users are responsible for the inherent limitations. However, the line between an addressable and inherent limitation here is not clear. Al Khourdajie gives the example of a simple factual errors as an addressable limitation, while factual errors are given as an example of an inherent limitation. This seems to indicate that the inherentness of a limitation is determined by the severity of the mistake being made by the model, but there is no guidance on how we might assess this severity. In the following sections I try to flag when a concern, such as data quality, are more explicitly the responsibility of developers, but often I am discussing both groups at once because I am not sure how to delineate these responsibilities.

⁶⁴ We may also argue, if we want to think about AI as an autonomous agent, that AI models themselves should assess values to make the correct recommendations. However, the first two chapters of this these

However, despite the recognized importance of good data in the field of AI, there is remarkably little research on the subject. A USA scientific roundtable on Data for AI gathered scientists from institutions like the National Institutes of Health (NIH) and National Science Foundation (NSF). This report found that in general there are weak AI methodologies and guidelines around data. The report says,

“Educated trial and error continues to guide advances in science applications of AI. Currently, there is no holistic theoretical approach unifying data, AI models, and the tasks performed by models that would help answer critical, foundational questions, such as: what information about a dataset can be deduced from a model trained on the data? Do models inherit the access limitations or classifications of the training data? For a given dataset and task, what are the best model, hyperparameters, and training method? When are more data needed, and how much incremental information will they have? Which data would make the biggest improvement? In what circumstances can a model be transferred to new data?” (Fagnan et al., 2019)

Although AI methodology has changed significantly since 2019, it is still common for quality data to be neglected. Qian et al. (2024) found that while data quality was unanimously the top priority of LLM developers, there is no consensus about what good data is or how to evaluate it.

This stands in stark contrast to the transparent scientific practices that Douglas (as well as scholars discussed in chapter 2) see as integrating values in a way that maintains objectivity, integrity, and trust. Scientific practices, although they are not perfect, have methods that make sense and use of human values. Medicine has rigorous requirements for the data and expected methodologies for both individual studies and meta-analyses. The IPCC has set procedures for achieving consensus. Our trust in scientific information comes, in part, from our belief in the methodology and process of science to make sense

provide reasoning for why we should think about AI models as evidence for, instead of replacements of, human judgement. Thus, this section treats AI models this way. I would also add that if we wanted to make arguments about AI as autonomous agents that can assess values, we would still have to satisfy the concerns about how their development introduces values brought up in this chapter, as well as further questions about their agency. For example, can AI models actually have opinions? This seems to not be the case at least for GPT-4o, which OpenAI rolled back because it was overly sycophantic, agreeing with whatever the user asked (OpenAI, 2025).

of human values in a way that is beneficial for knowledge production. Currently there is no comparable, agreed-upon processes for collection, treatment and assessment of AI data or models.

For AI models to match scientific epistemology we should expect their developers to have a firm justification (and ideally methodology) for the data that was used. For instance, AI models in radiology align with scientific goals and practices only because they can justify the use of medical images. The IPCC can use LLMs to search through scientific sources available online only because those are the sources we already expect to be searching through. Alternatively, when AI models do not have a firm justification for their data, we should expect them to perform poorly in decision making contexts, typically because of bias. As is the case with determining medical treatment, where we expect experts to consider a wide range of factors, and writing IPCC reports, where data would have to be representative of the opinions of the scientific community.

Further, we should expect some justification of the patterns AI models aim to predict. Scientific evidence can present predictions, but those predictions are reached by including a variety of perspectives. We know, through the process of deliberative acceptance and the diversity of scientists, that a variety of human values have been addressed in the scientific process. While AI data might contain a variety of perspectives, it is not clear how the model addresses these. Deep learning models find patterns that are opaque to us. This makes it difficult to make relevant value decisions about AI information because we are not always sure what patterns are “learned” by AI models. Thus, the AI methodology does not provide us with a way to assess the reliability of AI predictions.

To solve this problem, we may rely on tests that assess the accuracy of a model. For example, placing more trust in a model that makes fewer errors, trusting a model with 90% accuracy more than one with 70% accuracy. We may assume that the accuracy of the model will indicate the suitability of a model for predicting the underlying patterns. If it is unsuitable the model will perform poorly, if it is suitable the model will perform well. However, this makes it difficult to distinguish if poor performance is due to a poor model or unreliable patterns. As research into AI focuses more on the production of AI models than their underlying data, it is probable that this will lead to the assumption that

models simply need further development, rather than recognizing certain patterns may be unsuited for AI prediction.

Scientific methods are designed to establish the robustness of research results. Often with scientific experiments that aim to establish causal relations by counterfactual causal reasoning. This allows us to assess the strength of patterns of relation, and if we expect them to persist. While this does not solve the problem of inductive risk, it provides a better basis for us to assess the bounds of our predictions. It gives us insight beyond what is happening, allowing us to assess why something is happening, and how our actions might change what happens.⁶⁵ Thus, to trust AI models in decision making, we should expect some justification for why the patterns of the world it aims to predict are stable and useful. If we want to use AI models to aid decision-making, we cannot accept arguments that AI models can in principle predict shifts in patterns that we ourselves cannot foresee. If we argue for the adoption of AI, we must also bear the responsibility of proving that we can in practice make predictions of patterns that will occur.

This section has aimed to argue that AI models do not deal with human values in the same way scientific experts do. Scientific experts and processes attempt to account for human values; AI models obscure them. Not only do AI models make predictions from data in ways that are opaque to us, but the selection of data and the ways these models are used are often also opaque, unjustified, or ad hoc. What then, does this mean for expertise in decision making?

3.3 What is the Value of Human Expertise?

When we make decisions, especially important decisions, we often must consider a range of human values. This applies to medical decisions—especially treatment—where doctors must weigh disease severity against a patient's lifestyle and values. It applies to climate policy, where politicians must weigh available funding against public opinion and other factors. It even applies to scientists informing climate decisions, who must

⁶⁵ This is the difference between the scientific claim that people with asthma are at higher risk of negative health outcomes from pneumonia, and the AI claim that people with asthma are less likely to die from pneumonia.

determine what information is relevant to politicians, and how it should best be presented to a non-scientific audience.

Human values are not uniform, and how to deal with conflicting values is an ongoing challenge.⁶⁶ As discussed in chapter 2, the quality of scientific knowledge comes from transparent processes that help address human values and allow for communal evaluation of the correct decisions. While we may – as seen in chapter 1 – think of AI algorithms as having a similar methodology to science, this chapter has aimed to show that this is not the case. The development of AI does not include standard practices about the data involved in AI models, and there is no agreed upon way to assess how applicable an AI model is for a particular task. The production and use of AI algorithms face inductive risk, which requires addressing human values, but does not “deal with it” in the way science does.

We seem to be tempted to think of algorithms as neutral arbitrators. Our lives would be simpler if we could predict the world. If we could tell a computer everything and then know the correct course of action. However, many decisions are messy, and complex, and filled with human values. Even science must content with how it addresses human values. But science (ideally) has transparent institutional processes to cope with these value questions and give us reason to trust its knowledge. AI models, for a variety of reasons, feel scientific; in that they can provide us with expertise that is truthful, or accurate, or separate from human values. But AI models obscure human values rather than removing them. This chapter argued AI models obscure human values because it is difficult to determine the values shaping their creation. However, as discussed in chapter 2, this also depends on the scope in which AI is used. AI models obscure human values more severely when they replace processes that assess human values. This means that when we unthinkingly use AI to replace human decision-making not only are we producing systems with values we struggle to identify, but we are also using them to remove our own assertion of values.

⁶⁶ Who is making these value decisions is another question not fully explored in this thesis. There is a political dimension here where many value decisions in technology are made by software developers who are often male, white, and economically well off. This greatly shapes the values that are embedded in technology. As noted by Dr. D. Inkpen, the term “human values” may obscure the complexity of conflicting value assessments, as there is not one singular set of human values.

This is not to say that it is never possible for AI systems to assist in the production of useful knowledge. As we saw with diagnosis of rare disease, or finding sources for scientific assessments, AI systems can be useful to our decision making and the pursuit of science. Rather, the claim is that AI models themselves do not have ways to appropriately assess risk and human values – so AI production and use must manage them. If people want to use AI in decision-making contexts, the burden should fall to them to prove that the model is epistemically fit for the task. However, it seems too often the success of the AI models means they are applied to decision making contexts first and critically examined after. This is evidenced by biased AI models being used in a variety of contexts and applications, including but not limited to the medical system (Kumar et al., 2023), corporate tasks such as resume screening (Zomorodi, 2025), and the judicial system (Dressel and Hany, 2018).

AI has demonstrated the ability to complete a variety of tasks. AI models have beat experts in games of chess and go. Some LLMs are able to pass the bar exam in the 90th percentile (Sloan, 2023). Even in science, AI has played an increasingly important role. In 2024, both the physics and chemistry Nobel prizes were awarded for AI related achievements (Nobel Foundation, 2026a, 2026b). However, we must recognize that AI models are powerful pattern predictors, not arbitrators of an unknowable truth. The findings they have are contingent on the data they are provided, and the patterns they are trying to predict.

When we have to make decisions, it is easy to think that it may be resolved with enough information. That if the information is based in data, or correctly predicts patterns, we might know the right answer. But in many complex cases, a right answer depends on a number of human values. Computers cannot decide those values for us.

Bibliography

- Al-Antari, Mugahed A. 2023. Artificial intelligence for medical diagnostics—Existing and future AI technology! *Diagnostics* 13, 4: 688.
- Algaba, Andres, Vincent Holst, Floriano Tori, et al. 2025. “How Deep Do Large Language Models Internalize Scientific Literature and Citation Practices?” arXiv:2504.02767. Preprint, arXiv, April 3.
<https://doi.org/10.48550/arXiv.2504.02767>.
- Al Khourdjie, Alaa. 2025. “The Role of Artificial Intelligence in Climate Change Scientific Assessments.” *PLOS Climate* 4 (9): e0000706.
<https://doi.org/10.1371/journal.pclm.0000706>.
- American Medical Association. 2025. “Physician Sentiments Around the Use of AI in Health Care: Motivations, Opportunities, Risks, and Use Cases.” AMA Augmented Intelligence Research.
- Avanzo, Michele, Joseph Stancanella, Giovanni Pirrone, Annalisa Drigo, and Alessandra Retico. 2024. “The Evolution of Artificial Intelligence in Medical Imaging: From Computer Science to Machine and Deep Learning.” *Cancers* 16 (21): 3702.
<https://doi.org/10.3390/cancers16213702>.
- Balde, Gunjan, Soumyadeep Roy, Mainack Mondal, and Niloy Ganguly. 2025. “Evaluation of LLMs in Medical Text Summarization: The Role of Vocabulary Adaptation in High OOV Settings.” In *Findings of the Association for Computational Linguistics: ACL 2025*, edited by Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.findings-acl.1179>.
- Beatty, John, and Alfred Moore. 2010. “Should We Aim for Consensus?” *Episteme* 7 (3): 198–214. <https://doi.org/10.3366/epi.2010.0203>.
- Bender, Emily M., and Alex Hanna. 2025. *The AI Con: How to fight big tech’s hype and create the future we want*. Random House.
- Betz, Gregor. 2013. “In Defence of the Value Free Ideal.” *European Journal for Philosophy of Science* 3 (2): 207–20. <https://doi.org/10.1007/s13194-012-0062-x>.

- Bercovich, Eyal, and Marcia C. Javitt. 2018. "Medical Imaging: From Roentgen to the Digital Revolution, and Beyond." *Rambam Maimonides Medical Journal* 9 (4): e0034. <https://doi.org/10.5041/RMMJ.10355>.
- Bridle, James. 2024. "So, Amazon's 'AI-Powered' Cashier-Free Shops Use a Lot of ... Humans. Here's Why That Shouldn't Surprise You." *Opinion. The Guardian*, April 10. <https://www.theguardian.com/commentisfree/2024/apr/10/amazon-ai-cashier-less-shops-humans-technology>.
- Briggle, Adam. 2024. *A Field Guide to Climate Change: Understanding the Problems*. Broadview Press.
- Bouguettaya, Ayoub, Elizabeth M. Stuart, and Elias Aboujaoude. 2025. "Racial Bias in AI-Mediated Psychiatric Diagnosis and Treatment: A Qualitative Comparison of Four Large Language Models." *Npj Digital Medicine* 8 (1): 332. <https://doi.org/10.1038/s41746-025-01746-4>.
- Buck, Holly Jean, Alaa Al Khourdajie, Shinichiro Asayama, and Oliver Geden. 2025. "Four Scenarios for an IPCC Navigating Artificial Intelligence." *SWP Working Papers*. https://www.swp-berlin.org/publications/products/arbeitspapiere/WP_Geden_etal_IPCC_290925.pdf
- Buolamwini, Joy Adowaa. 2017. *Gender Shades: Intersectional Phenotypic and Demographic Evaluation of Face Datasets and Gender Classifiers*.
- Butler, Patrick. 2025. N.L.'s 10-year education action plan cites sources that don't exist. *CBC*, <https://www.cbc.ca/news/canada/newfoundland-labrador/education-accord-nl-sources-dont-exist-1.7631364>
- Cartwright, Nancy. 2011. "A Philosopher's View of the Long Road from RCTs to Effectiveness." *The Lancet* 377 (9775): 1400–1401. [https://doi.org/10.1016/S0140-6736\(11\)60563-1](https://doi.org/10.1016/S0140-6736(11)60563-1).
- Chavelli, Felix, and Sarah Connors. 2023. *Analysis of the WGI Contribution to the Sixth Assessment Report: Review of the WGI AR6 References*. Zenodo. <https://doi.org/10.5281/zenodo.7615825>.
- Cheval, Sorin, Vlad-Alexandru Amihăesei, Zenaida Chitu, et al. 2024. "A Systematic Review of Urban Heat Island and Heat Waves Research (1991–2022)." *Climate*

- Risk Management 44 (January): 100603.
<https://doi.org/10.1016/j.crm.2024.100603>.
- Collins, Harry. 2014. *Are we all scientific experts now?* John Wiley & Sons.
- Conway, Erik M., and Naomi Oreskes. 2010. *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Climate Change*. Bloomsbury.
- Douglas, Heather. 2000. "Inductive Risk and Values in Science." *Philosophy of Science* 67 (4): 559–79. <https://doi.org/10.1086/392855>.
- Dressel, Julia, and Hany Farid. 2018. "The Accuracy, Fairness, and Limits of Predicting Recidivism." *Science Advances* 4 (1): eaao5580.
<https://doi.org/10.1126/sciadv.aao5580>.
- Educating Silicon. 2024. "How Much LLM Training Data Is There, in the Limit?" Accessed March 14, 2026. <https://www.educatingsilicon.com/2024/05/09/how-much-llm-training-data-is-there-in-the-limit/>.
- Erickson, Bradley J. and Brian Bartholmai. 2002. "Computer-Aided Detection and Diagnosis at the Start of the Third Millennium." *Journal of Digital Imaging* 15 (2): 59–68. <https://doi.org/10.1007/s10278-002-0011-x>.
- Farr, Deonna E., Thad Benefield, Mi Hwa Lee, Essie Torres, and Louise M. Henderson. 2024. "Multilevel Contributors to Racial and Ethnic Inequities in the Resolution of Abnormal Mammography Results." *Cancer Causes & Control : CCC* 35 (7): 995–1009. <https://doi.org/10.1007/s10552-024-01851-x>.
- Fay, David S., and Ken Gerow. 2018. "A Biologist's Guide to Statistical Thinking and Analysis." In *WormBook: The Online Review of C. Elegans Biology* [Internet]. WormBook. <https://www.ncbi.nlm.nih.gov/books/NBK153593/>.
- Ferrario, Andrea, Alessandro Facchini, and Alberto Termine. 2024. Experts or authorities? The strange case of the presumed epistemic superiority of artificial intelligence systems. *Minds and Machines* 34, 3: 30.
- Flemyng, Ella, Anna Noel-Storr, Biljana Macura, et al. 2025. Position Statement on Artificial Intelligence (AI) Use in Evidence Synthesis across Cochrane, the Campbell Collaboration, JBI and the Collaboration for Environmental Evidence

- 2025 | Cochrane Library. Accessed November 11, 2025.
<https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.ED000178/full>.
- Ford, James, Robbert Biesbroek, Lea Berrang-Ford, et al. 2025. “Recommendations for Producing Knowledge Syntheses to Inform Climate Change Assessments.” *Nature Climate Change* 15 (June). <https://doi.org/10.1038/s41558-025-02354-6>.
- Galison, Peter. 1999. “Trading Zone: Coordinating Action and Belief (1998 Abridgment).” In *The Science Studies Reader*, edited by Mario Biagioli. Routledge.
- Gartner. 2026. “Gartner Says Worldwide AI Spending Will Total \$2.5 Trillion in 2026.” Accessed March 14, 2026. <https://www.gartner.com/en/newsroom/press-releases/2026-1-15-gartner-says-worldwide-ai-spending-will-total-2-point-5-trillion-dollars-in-2026>.
- Giouroukou, Konstantina, Kostas Marias, Manolis Tsiknakis, and Michail E. Klontzas. 2026. “Rethinking Privacy in Medical Imaging AI: From Metadata and Pixel-Level Identification Risks to Federated Learning and Synthetic Data Challenges.” *Radiology. Artificial Intelligence* 8 (1): e250273. <https://doi.org/10.1148/ryai.250273>.
- Gomstyn, Alice, and Alexandra Jonker. n.d. “What Is Model Collapse? | IBM.” Accessed March 14, 2026. <https://www.ibm.com/think/topics/model-collapse>.
- Göndöcs, Dóra, and Viktor Dörfler. 2024. “AI in Medical Diagnosis: AI Prediction & Human Judgment.” *Artificial Intelligence in Medicine* 149 (March): 102769. <https://doi.org/10.1016/j.artmed.2024.102769>.
- Goyal, Navita, Connor Baumler, Tin Nguyen, and Hal Daumé III. 2024. “The Impact of Explanations on Fairness in Human-AI Decision-Making: Protected vs Proxy Features.” *Proceedings of the 29th International Conference on Intelligent User Interfaces* (New York, NY, USA), IUI ’24, April 5, 155–80. <https://doi.org/10.1145/3640543.3645210>.
- Green, John. 2025. *Everything Is Tuberculosis: The History and Persistence of Our Deadliest Infection*. Crash Course Books, an imprint of Penguin Random House LLC.

- Hawkins, Kaitlyn, Rui Lui, and Molly Rivers. 2021. The Researchers That Search Engines Make Invisible - CLEAR - Civic Laboratory for Environmental Action Research. Blog. June 10. <https://civiclaboratory.nl/2021/06/10/the-researchers-that-search-engines-make-invisible/>.
- Hayes, Leyla. 2025. AI used for skin cancer checks at London hospital. BBC. <https://www.bbc.com/news/articles/czd3ygd7mrno>
- Henderson, Leah. 2024. "The Problem of Induction." In The Stanford Encyclopedia of Philosophy, Winter 2024, edited by Edward N. Zalta and Uri Nodelman. Metaphysics Research Lab, Stanford University. <https://plato.stanford.edu/archives/win2024/entries/induction-problem/>.
- Hussein, Mohamed A. n.d. "Visualising AI Spending: How Does It Compare with History's Mega Projects?" Al Jazeera. Accessed March 14, 2026. <https://www.aljazeera.com/news/2026/2/19/visualising-ai-spending-how-does-it-compare-with-historys-mega-projects>.
- IBM Data and AI Team. 2023. "AI vs. Machine Learning vs. Deep Learning vs. Neural Networks" IBM, July 6. <https://www.ibm.com/think/topics/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks>.
- IPCC, thirty-seventh session. 2013a, October 14-18. "Principles Governing IPCC Work." <https://www.ipcc.ch/site/assets/uploads/2018/09/ipcc-principles.pdf>.
- IPCC, thirty-seventh session. 2013b, October 14-18. "Procedures for the Preparation, Review, Acceptance, Adoption, Approval, and Publication of IPCC Reports: Appendix A to the Principles Governing IPCC Work." <https://www.ipcc.ch/site/assets/uploads/2018/09/ipcc-principles-appendix-a-final.pdf>.
- IPCC. 2021. "Box 1.1, Figure 1: The IPCC AR6 approach for characterizing understanding and uncertainty in assessment findings" in Climate Change 2021: The Physical Science Basis. Intergovernmental Panel on Climate Change. Cambridge University Press. 10.1017/9781009157896.003.
- IPCC. 2024b, August 20. "Artificial Intelligence Support Officer in the IPCC Working Group II Technical Support Unit" (job posting). Intergovernmental Panel on

- Climate Change. <https://www.ipcc.ch/2024/08/20/ai-support-officer-ipcc-wg-ii-tsu/>.
- IPCC. 2025a. “About the IPCC.” Intergovernmental Panel on Climate Change. Accessed October 14, 2025. <https://www.ipcc.ch/about/>.
- IPCC. 2025b. “Synthesis Report” (introductory webpage). Intergovernmental Panel on Climate Change. Accessed October 14, 2025. <https://www.ipcc.ch/synthesis-report/>.
- IPCC. 2025c. “The Task Force on National Greenhouse Gas Inventories.” Intergovernmental Panel on Climate Change. Accessed October 14, 2025. <https://www.ipcc.ch/working-group/tfi/>.
- Jancer, Matt. 2025. “Google Says Gmail Doesn’t Use Your Emails to Train Its AI, After All.” VICE, November 24. <https://www.vice.com/en/article/google-says-gmail-doesnt-use-emails-to-train-ai/>.
- Kalai, Adam Tauman, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. 2025. “Why Language Models Hallucinate.” arXiv:2509.04664. Preprint, arXiv, September 4. <https://doi.org/10.48550/arXiv.2509.04664>.
- Kitcher, Philip. 2011. Science in a democratic society. Poznan Studies in the Philosophy of the Sciences & the Humanities 101.
- Lai, Vivian, Chacha Chen, Alison Smith-Renner, Q. Vera Liao, and Chenhao Tan. 2023. Towards a science of human-AI decision making: An overview of design space in empirical human-subject studies. In Proceedings of the 2023 ACM conference on fairness, accountability, and transparency, pp. 1369-1385.
- Lee, Jisoo, Jieun Lee, and Jeong-Ju Yoo. 2025. “The Role of Large Language Models in the Peer-Review Process: Opportunities and Challenges for Medical Journal Reviewers and Editors.” Journal of Educational Evaluation for Health Professions 22 (January): 4. <https://doi.org/10.3352/jeehp.2025.22.4>.
- Lewis, Patrick, Ethan Perez, Aleksandra Piktus, et al. 2021. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” arXiv:2005.11401. Preprint, arXiv, April 12. <https://doi.org/10.48550/arXiv.2005.11401>.

- Lodwick, Gwilym S., Arch H. Turner, Lee B. Lusted, and Arch W. Templeton. 1966. "Computer-Aided Analysis of Radiographic Images." *Journal of Chronic Diseases* 19 (4): 485–96. [https://doi.org/10.1016/0021-9681\(66\)90122-6](https://doi.org/10.1016/0021-9681(66)90122-6).
- Longino, Helen E. 1990. *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton University Press. <https://doi.org/10.2307/j.ctvx5wbfbz>.
- Louro, Celeste Rodriguez. 2025. "AI Systems Are Built on English – but Not the Kind Most of the World Speaks." *The Conversation*, May 5. <https://doi.org/10.64628/AA.4mhwhkhr5>.
- Li, Kangming, Daniel Persaud, Kamal Choudhary, Brian DeCost, Michael Greenwood, and Jason Hattrick-Simpers. 2023. "Exploiting Redundancy in Large Materials Datasets for Efficient Machine Learning with Less Data." *Nature Communications* 14 (1): 7283. <https://doi.org/10.1038/s41467-023-42992-y>.
- Lusted, Lee B. 1955. "Medical Electronics." *New England Journal of Medicine* 252 (14): 580–85. <https://doi.org/10.1056/NEJM195504072521405>.
- Mansour, Sahar, Somia Soliman, Abisha Kansakar, Ahmed Marey, Christiane Hunold, and Mennatallah Mohamed Hanafy. 2022. "Strengths and Challenges of the Artificial Intelligence in the Assessment of Dense Breasts." *BJR Open* 4 (1): 20220018. <https://doi.org/10.1259/bjro.20220018>.
- McCann, S.B., and R.B. Taylor. "Coast." *The Canadian Encyclopedia*. Historica Canada. Article published March 29, 2009; Last Edited December 10, 2024.
- McCulloch, Gretchen. 2020. *Because Internet: Understanding the New Rules of Language*. First Riverhead trade paperback edition. Riverhead Books.
- Melumad, Shiri, and Jin Ho Yun. 2025. "Experimental Evidence of the Effects of Large Language Models versus Web Search on Depth of Learning." *PNAS Nexus* 4 (10): pgaf316. <https://doi.org/10.1093/pnasnexus/pgaf316>.
- Messeri, Lisa, and Molly J. Crockett. 2024. Artificial intelligence and illusions of understanding in scientific research. *Nature* 627, 8002:49-58.
- Mishra, Tanisha, Edward Sutanto, Rini Rossanti, et al. 2024. "Use of Large Language Models as Artificial Intelligence Tools in Academic Research and Publishing

- among Global Clinical Researchers.” *Scientific Reports* 14 (1): 31672.
<https://doi.org/10.1038/s41598-024-81370-6>.
- Moore, Alfred. 2017. *Critical elitism: Deliberation, democracy, and the problem of expertise*. Cambridge University Press.
- Moore, James H. 1985. Are there decisions computers should never make. *Ethical issues in the use of computers* 120:130.
- Mosolova, Anna, Marie Candito, and Carlos Ramisch. 2025. “In the LLM Era, Word Sense Induction Remains Unsolved.” In *Findings of the Association for Computational Linguistics: ACL 2025*, edited by Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.findings-acl.882>.
- Nichols, Thomas M. 2024. *The death of expertise: The campaign against established knowledge and why it matters*. Oxford University Press.
- Nobel Foundation. 2026a. "The Nobel Prize in Chemistry 2024." NobelPrize.org. Accessed March 23, 2026.
<https://www.nobelprize.org/prizes/chemistry/2024/summary/>.
- Nobel Foundation. 2026b. "The Nobel Prize in Physics 2024." NobelPrize.org. Accessed March 23, 2026. <https://www.nobelprize.org/prizes/physics/2024/summary/>.
- OpenAI. 2025. “Sycophancy in GPT-4o: What Happened and What We’re Doing about It.” OpenAI, April 29. <https://openai.com/index/sycophancy-in-gpt-4o/>.
- Oppenheimer, Michael, Naomi Oreskes, Dale Jamieson, et al. 2019. *Discerning Experts: The Practices of Scientific Assessment for Environmental Policy*. University of Chicago Press.
- Page, Matthew J., Joanne E. McKenzie, Patrick M. Bossuyt, et al. 2021. “The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews.” *Research Methods & Reporting*. *BMJ* 372 (March): n71.
<https://doi.org/10.1136/bmj.n71>.
- Peterson, Andrew J. 2024. “AI and the Problem of Knowledge Collapse.” arXiv.
<https://doi.org/10.48550/arXiv.2404.03502>

- Peters, Uwe, and Benjamin Chin-Yee. 2025. "Generalization Bias in Large Language Model Summarization of Scientific Research." *Royal Society Open Science* 12 (4): 241776. <https://doi.org/10.1098/rsos.241776>.
- Pielke Jr, Roger A. 2007. *The honest broker: Making sense of science in policy and politics*. Cambridge University Press.
- Punnath, Aswaj, Sahanashree Ramakrishnaiah, Udayakantha Warnakulasuriyage Sudesh, Aniruddha Marathe, and Priyadarsanan Dharma Rajan. 2021. "Two New Species of Doryline Ants (Hymenoptera, Formicidae) with 11-Segmented Antennae from India." *ZooKeys* 1056 (January): 59–72. <https://doi.org/10.3897/zookeys.1056.68722>.
- Qian, Crystal, Emily Reif, and Minsuk Kahng. 2024. "Understanding the Dataset Practitioners Behind Large Language Models." *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (New York, NY, USA), CHI EA '24, May 11, 1–7. <https://doi.org/10.1145/3613905.3651007>.
- Rahman, Robi, and David Owen. n.d. "The Size of Datasets Used to Train Language Models Doubles Approximately Every Six Months." *Epoch AI*. Accessed March 14, 2026. <https://epoch.ai/data-insights/dataset-size-trend>.
- Richens, Jonathan G., Ciarán M. Lee, and Saurabh Johri. 2020. "Improving the Accuracy of Medical Diagnosis with Causal Machine Learning." *Nature Communications* 11 (1): 3923. <https://doi.org/10.1038/s41467-020-17419-7>.
- Salesforce. 2024. "Just 11% of CIOs Have Fully Implemented AI as Data and Security Concerns Hinder Adoption." 2024. Salesforce, October 15. <https://www.salesforce.com/news/stories/cio-ai-trends/>.
- Sarewitz, Daniel. 2004. How science makes environmental controversies worse. *Environmental science & policy* 7, 5: 385-403.
- Schaffer, James, John O'Donovan, James Michaelis, Adrienne Raglin, and Tobias Höllerer. 2019. I can do better than your AI: expertise and explanations. In *Proceedings of the 24th international conference on intelligent user interfaces*, pp. 240-251.

- Schoenthaler, Antoinette, and Natasha Williams. 2022. "Looking Beneath the Surface: Racial Bias in the Treatment and Management of Pain." *JAMA Network Open* 5 (6): e2216281. <https://doi.org/10.1001/jamanetworkopen.2022.16281>.
- Schraagen, Jan Maarten, and Jurriaan van Diggelen. 2021. A brief history of the relationship between expertise and artificial intelligence. In *Expertise at work: Current and emerging trends*, pp. 149-175. Cham: Springer International Publishing.
- Schuemie, Martijn J., Jan A. Kors, and Barend Mons. 2005. "Word Sense Disambiguation in the Biomedical Domain: An Overview." *Journal of Computational Biology* 12 (5): 554–65. <https://doi.org/10.1089/cmb.2005.12.554>.
- Sennrich, Rico, Barry Haddow, and Alexandra Birch. 2016. "Neural Machine Translation of Rare Words with Subword Units." In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, edited by Katrin Erk and Noah A. Smith. Association for Computational Linguistics. <https://doi.org/10.18653/v1/P16-1162>.
- Singh Chawla, Dalmeet. 2024. "The Citation Black Market: Schemes Selling Fake References Alarm Scientists." *Nature* 632 (8027): 966–966. <https://doi.org/10.1038/d41586-024-01672-7>.
- Sloan, Karen. 2023. "Bar Exam Score Shows AI Can Keep up with 'human Lawyers,' Researchers Say." *Technology*. Reuters, March 15. <https://www.reuters.com/technology/bar-exam-score-shows-ai-can-keep-up-with-human-lawyers-researchers-say-2023-03-15/>.
- Stanford HAI. 2026. "Chapter 4 Economy | The 2025 AI Index Report". Accessed March 14, 2026. <https://hai.stanford.edu/ai-index/2025-ai-index-report/economy>.
- Stegenga, Jacob. 2009. Robustness, discordance, and relevance. *Philosophy of Science* 76, 5: 650-661.
- Tegmark, Max. 2008. "The Mathematical Universe." *Foundations of Physics* 38 (2): 101–50. <https://doi.org/10.1007/s10701-007-9186-9>.
- Trafton, Anne, 2024. "Study Reveals Why AI Models That Analyze Medical Images Can Be Biased." June 28. MIT news. <https://news.mit.edu/2024/study-reveals-why-ai-analyzed-medical-images-can-be-biased-0628>.

- U.S. Food and Drug Administration. 1998. "Summary of Safety and Effectiveness Data: ImageChecker M1000 System." Premarket Approval Application P970058. Rockville, MD: Center for Devices and Radiological Health.
- Van Noorden, Richard, and Jeffrey M. Perkel. 2023. AI and science: what 1,600 researchers think. *Nature* 621, 7980: 672-675.
- Villalobos, Pablo, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. 2024. "Will We Run out of Data? Limits of LLM Scaling Based on Human-Generated Data." arXiv:2211.04325. Preprint, arXiv, June 4. <https://doi.org/10.48550/arXiv.2211.04325>.
- Wagner, Niklas, and Anna-Katharina Hornidge. 2025. "Unlearning Modernity? A Critical Examination of the Intergovernmental Panel on Climate Change (IPCC)." *Climatic Change* 178 (2): 32. <https://doi.org/10.1007/s10584-025-03866-y>.
- WHO. n.d. "Social Determinants of Health." World Health Organization. Accessed March 27, 2026. <https://www.who.int/health-topics/social-determinants-of-health>.
- Wu, Jinyang, Feihu Che, Xinxin Zheng, et al. 2024. "Can Large Language Models Understand Uncommon Meanings of Common Words?" arXiv:2405.05741. Preprint, arXiv, May 10. <https://doi.org/10.48550/arXiv.2405.05741>.
- Xiao, Tianqi, Francesco Fuso Nerini, H. Damon Matthews, Massimo Tavoni, and Fengqi You. 2025. "Environmental Impact and Net-Zero Pathways for Sustainable Artificial Intelligence Servers in the USA." *Nature Sustainability*, November 10, 1–13. <https://doi.org/10.1038/s41893-025-01681-y>.
- Xu, Yongjun, Xin Liu, Xin Cao, Changping Huang, Enke Liu, Sen Qian, Xingchen Liu et al. 2021. Artificial intelligence: A powerful paradigm for scientific research. *The Innovation* 2, 4.
- Zheng, Jonathan, Alan Ritter, and Wei Xu. 2024. "NEO-BENCH: Evaluating Robustness of Large Language Models with Neologisms." In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, edited by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.acl-long.749>.
- Zöller, Nikolas, Julian Berger, Irving Lin, Nathan Fu, Jayanth Komarneni, Gioele Barabucci, Kyle Laskowski et al. 2025. Human–AI collectives most accurately

diagnose clinical vignettes. *Proceedings of the National Academy of Sciences* 122, 24: e2426153122.

Zomorodi, Manoush. 2025. "AI Is Screening Your Resume. Here's How to Make It Past the Bots." NPR, October 3. <https://www.npr.org/2025/10/03/nx-s1-5534959/are-ai-hiring-tools-any-good-this-journalist-found-widespread-bias-and-bugs>.